

Learned representations and kernel corrections for radiative-transfer emulation

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Abstract

We compare neural networks, exact Matérn regression and their combinations on a six-dimensional radiative-transfer table over the 285-band EMIT grid. Across ten partitions with 18,884 training samples and validation-based selection, an input-scaled kernel gives 0.095% mean relative radiance error, compared with 0.760% for cubic regression and 0.387% for a neural network. Residual correction reduces the network error to 0.141%; a kernel on wider-network features reaches 0.079%. The wide convex stack improves the mean radiance error by only 0.0006 percentage points while increasing the mean within-split 95th-percentile reflectance error from 7.57 to 22.71 points, with a larger tail at every split. For a physically constrained retrieval, we prove a constant-one bound by the transmission-weighted component error divided by the transmission. We give a sharp worst-case transfer exponent under a transmission-tail hypothesis and conditional-error conditions that retain the component-error rate. We also distinguish the raw-error quadratic used to fit retrieval-weighted stacks from the projected, nonlinear retrieval used to score them. The experiments concern interpolation of one adaptively explored table: overlapping partitions and finite-range transmission profiles do not establish independent validation or asymptotic learning rates. Supplementary OCO-2 and multi-fidelity comparisons illustrate how the evaluation metric and input representation change the relative performance of the specified model families.

1 Introduction

Atmospheric correction for imaging spectroscopy uses radiative-transfer calculations to relate atmospheric and viewing conditions to measured radiance. Learned emulators replace repeated simulator evaluations with a fitted map to spectral coefficients. Neural emulators have been incorporated into reflectance-retrieval systems [1, 2]; Gaussian-process emulators have also been developed for the OCO-2 forward model [3]. Physics-guided multi-fidelity models use lower-fidelity calculations as additional information [4]. These approaches motivate two distinct questions: which representation gives an accurate forward emulator, and which errors matter when that emulator is used in an inverse calculation?

We study direct kernel regression, neural networks and two post-hoc combinations. In residual correction, the network supplies a mean and a kernel fits its training residual. In feature regression, the network supplies a frozen representation and a kernel fits the target on that representation [12]. These constructions differ from jointly trained deep kernel learning [10]

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and from regularizing intermediate network layers with kernel flows [11]. Their comparison complements work on kernel competitors for operator learning [5], neural representations such as DeepONet and Fourier neural operators [6, 7], and classical kernel regression [8].

The principal experiment uses 23,313 tabulated radiative-transfer evaluations on the 285-band EMIT wavelength grid [13]. The input has six coordinates and the standardized outputs have a small principal-component representation, making a dense kernel solve feasible at the full training size. Ten common partitions compare the model families. Additional experiments vary training size, output rank, input scaling, network width and kernel selection. The input-scaled kernel is substantially more accurate than the isotropic kernel. A wider network has a larger standalone error under the tested training configuration but supplies features on which a kernel predicts more accurately. These findings concern the fitted implementations and budgets, not an intrinsic ordering of kernels and networks.

The central empirical result is a forward–inverse reversal. The wide stack’s mean radiance improvement over its feature kernel is 0.0006 percentage points, whereas its mean within-split 95th-percentile reflectance error rises from 7.57 to 22.71 points. The reflectance tail increases at all ten partitions. These are all-band diagnostics at surface reflectance 0.7, including poorly conditioned absorption entries; they are not estimates of operational retrieval accuracy. The result shows that a small forward gain can accompany a substantial change in an inverse-error tail.

Our analysis separates three levels of explanation. Exact finite-perturbation identities relate radiance and reflectance errors through the inverse denominator. A constant-one transfer inequality controls a physically constrained retrieval by component error relative to transmission. Population tail and conditional-error assumptions then yield rate bounds, including a sharp worst-case exponent for this inverse. The transmission profiles measured on a finite table are diagnostics of the error-amplification mechanism, not verification of those assumptions arbitrarily close to zero. For residual correction, a separate Hilbert-space identity gives the exact error-alignment criterion, while a PCA decomposition keeps truncation error distinct from kernel-regression error.

The stacking analysis is implementation-specific. We distinguish a total-flux quadratic from the separable direct- and diffuse-flux quadratic used by the diagnostic driver. Projection is applied to the summed flux during retrieval scoring, not inside the fitted quadratic. The latter therefore remains convex, but the numerical weighting routine is an approximation to simplex-constrained least squares rather than a certified exact optimizer. The retrieval bound holds for any feasible weights on its stated physical domain; it does not assert that minimizing a validation surrogate improves every test metric.

Scope of the evidence. The main comparisons have complete per-split summaries, but model development included inspection of test results from this same table. The overlapping partitions quantify within-table split sensitivity rather than ten independent confirmation experiments. Secondary evidence has unequal replication and archival completeness. We distinguish recomputable summaries, retained aggregate tables and experiments requiring unavailable sample-level arrays. The published benchmark comparison uses the benchmark’s stated metrics and archived scores; it is not a retraining of its reference model under a matched computational budget.

Section 2 reviews the relevant constructions, and Section 3 specifies the data and evaluation protocol. Sections 4–6 give the models, primary comparison and ablations. Sections 7 and 8 develop the residual and retrieval analyses. Sections 9–12 report the associated diagnostics and supplementary comparisons; Sections 13 and 14 discuss OCO-2 and ten further configurations.

The discussion states the implications and remaining reproducibility limits. Proofs and extended diagnostic tables are collected in the appendices.

2 Related work

Radiative transfer and its surrogates. Relevant radiative-transfer codes include 6S [14], libRadtran [15, 16] and MODTRAN [17]; lookup-table calculations enter atmospheric correction [2] and optimal estimation [18]. These references describe the surrounding methodology, not a recovered generation configuration for the EMIT arrays used here. The carbon-dioxide product of OCO-2 [19, 20] rests on the forward model and inverse method of [21, 22]. Surrogates enter because the forward model is the cost. Neural emulation of longwave radiation goes back to [23] and was carried into climate models by [24]; later work measures how accurate such emulators can be made [25, 26] and how learned parameterizations behave when the host model is stepped forward [27, 28]. In imaging spectroscopy and land-surface inference the same idea appears as emulator toolboxes for radiative-transfer models [29, 30], Gaussian-process emulation for parameter retrieval [31], and the wider use of Gaussian processes in earth observation [32]; [1] places an emulator inside a reflectance retrieval and [3] fits a cross-validated Gaussian process to the OCO-2 forward model. Replacing an expensive simulator by a statistical model is a classical problem [33], as is using a cheap model to inform an expensive one [34, 35, 36], surveyed for propagation and inference in [37]; the network of [4] is a multi-fidelity instance on the corpus rescored in Section 12. The measurement requirements behind the tables used here are those of the EMIT mission [13] and of the surface-imaging missions that follow it [38].

Kernel regression. The estimators are classical: interpolation in a reproducing-kernel Hilbert space [39], smoothing splines and the representer theorem [40, 41, 42], ridge regularization [43] read as spectral filtering of an ill-posed problem [44], and the Matérn family, whose smoothness index fixes the differentiability of the sample paths [45, 8]. Rates under source and capacity conditions are given in [46, 47, 48, 49]; the textbook accounts are [50, 51, 52]. Each component here is a vector of principal coefficients, so the multi-output formulation is the relevant one [53, 54]. Kernels also serve as solvers rather than regressors: Bayesian numerical homogenization [55], operator-adapted wavelets [56] and the Gaussian-process treatment of nonlinear partial differential equations [57] use the same estimator, and cross-validating the kernel itself gives the kernel-flow family [58, 59, 11].

Cost. An exact solve is cubic in the number of training rows, which has produced a long line of approximations: the Nyström method and its analysis [60, 61, 62], random features [63] and their operator-learning counterpart [64], preconditioned solvers at large scale [65, 66], inducing-point and variational Gaussian processes [67, 68, 69], and structured interpolation with hardware-aware linear algebra [70, 71]. At $n = 18,884$ the exact factorization is affordable, so approximation enters this study only as a perturbation whose size is measured (Section 9).

Choosing and combining kernels. A kernel can be built rather than picked: spectral mixtures [72], additive structure [73], compositional search over a grammar [74], and per-coordinate relevance fitted by marginal likelihood [8] or by cross-validation [58]. Several kernels can be combined instead of one being chosen, by semidefinite programming [75], by the algorithms surveyed in [76], or by alignment with the target Gram matrix [77, 78]. Combining predictors

rather than kernels is older still [79, 80, 81]. Within each run, selection uses validation rows [82, 83, 84]. Section 8.2 gives a finite-class guarantee under a stronger independent-selection protocol, which the reused validation sets in this study do not satisfy.

Networks, and the kernels they induce. Feedforward networks are universal approximators [85]; those trained here use the standard adaptive optimizer [86] with decoupled weight decay [87] and early stopping [88]. Dropout is another established regularizer [89], but is not an arm of the reported regularization table. In the infinite-width limit a network is a Gaussian process [90, 91, 92] whose kernel is the arc-cosine kernel of [93], and gradient descent on a wide network follows the neural tangent kernel [94, 95]; both enter the kernel comparison of Section 10 as heads. The limit is not the whole story, since a finite network learns its representation [97], and the comparison between kernel machines and trained networks is itself unsettled [96]; a learned representation under a kernel is what deep kernel learning [10], the Gaussian process with a neural-operator mean [9] and the construction of [12] exploit. Kolmogorov–Arnold networks [98, 99] are the architecture behind the multi-fidelity model of [4]. Operator learning supplies the other comparison: DeepONet [6], Fourier neural operators [7] and the general formulation [100], with fair-comparison protocols [101], physics-informed training [102], the cost-accuracy trade-off [103], kernel competitors [5], and the benchmark collections [104, 105, 106] from which Section 14 draws.

3 Data and protocol

The data are $n = 23,313$ tabulated radiative-transfer evaluations. Each input $x \in \mathbb{R}^6$ collects the atmospheric and viewing state; the coordinate ranges are the ranges over which such lookup tables are built for imaging-spectroscopy atmospheric correction [2, 17] — an aerosol optical depth in $[0.05, 1.2]$, a surface elevation in $[-0.5, 6]$ km, a water-vapor column in $[0.05, 7]$ cm, a relative azimuth in $[0, 180]^\circ$, a solar zenith angle in $[0, 65]^\circ$, and a view zenith angle within 30° of nadir. Each output is a set of four spectral components on the same 285-point EMIT wavelength grid spanning 381–2493 nm: path radiance Y_1 , direct flux Y_2 , diffuse flux Y_3 , and spherical albedo Y_4 (Figure 1). The quantities retrieved downstream are the radiance at a test surface reflectance ρ ,

$$L(\rho) = Y_1 + \rho \frac{Y_2 + Y_3}{1 - \rho Y_4}, \quad (1)$$

and the reflectance recovered by inverting (1) with predicted components at the true radiance,

$$\hat{\rho} = \frac{L - \hat{Y}_1}{\hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)}, \quad (2)$$

both evaluated at $\rho = 0.7$ in the primary comparison. The numerical coordinate ranges above are the reported lookup-table description, not independently recovered sample extrema: the public records do not include the raw arrays or simulator-generation configuration.

The protocol is the same for every family. The table is split into a test block of 2,331 rows and a remaining block of 20,982; 2,098 of the latter are held out for validation, leaving 18,884 for fitting. The procedure is repeated with seeds 101–110. Standardization and principal components are fitted only on each split’s training rows. Hyperparameters, checkpoints, coordinatewise selections and combination weights use that split’s validation rows. Each fitted family is then

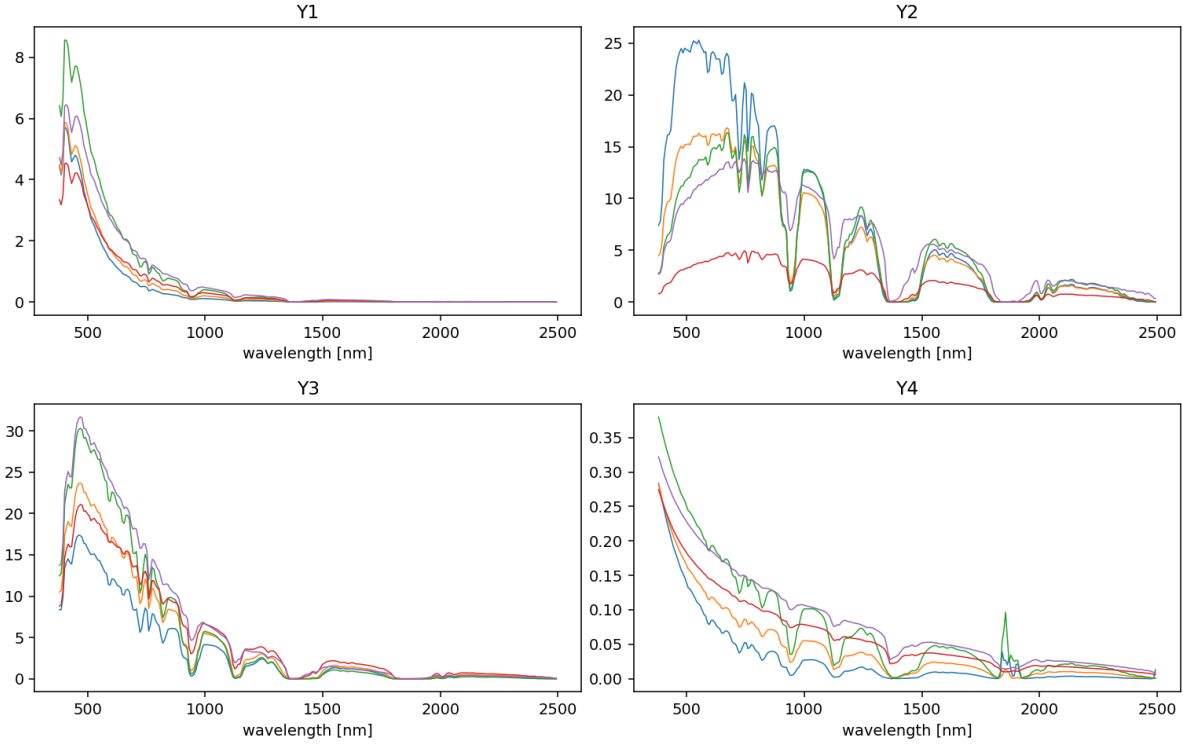


Figure 1: Example output curves across the EMIT wavelength grid, five samples per panel. The curves are smooth, share the deep water-vapor absorption features near 1380 and 1900 nm, and differ mainly by smooth rescalings along the state axes.

evaluated on that split’s test rows. Unless stated otherwise, tables report the mean and sample standard deviation over the ten splits.

The model families and ablations were chosen during exploratory analysis of the same table, including inspection of test-set results. Repeated partitions therefore measure sensitivity to the split within this table, rather than independent external validation. Their test sets overlap, and their standard deviations are descriptive, not confidence intervals from ten independent datasets. Section 5.1 reports paired differences on common splits. Validation rows are shared by early stopping, tuning and model combination. The primary model comparison uses $\rho = 0.7$. Supplementary stored retrieval diagnostics also evaluate a prescribed reflectance grid, but no EMIT atmospheric states outside the table or independent operational retrievals are tested.

Three empirical structural diagnostics motivate the model comparison.

The outputs are low rank. On standardized training outputs, 99.99% of the variance of every component lies in the first 11–25 principal components, and 99.9999% in at most 44 (Figure 2, left); the minimum adjacent-band correlation across the four components is 0.994952 on the training rows. All regressors below act on the first 64 principal coefficients, whose training unexplained-variance fraction is below 10^{-6} , except the networks, which are trained directly on the 285 standardized bands. A small aggregate variance tail is not a uniform relative-error guarantee in physical units; Section 7 states the projection error explicitly.

The measured input relevance is anisotropic. Shuffling the aerosol coordinate reduces a linear model’s validation R^2 by 0.89; the corresponding reduction is 0.54 for solar zenith and

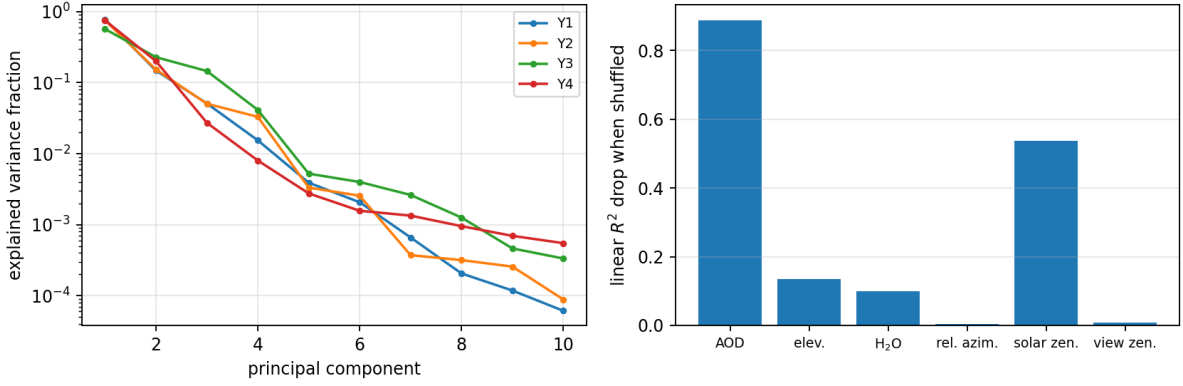


Figure 2: Left: explained-variance spectra of the standardized outputs. Every component concentrates 99.99% of its variance in 11–25 principal components out of 285 bands. Right: drop in linear validation R^2 when a single standardized input coordinate is shuffled. Aerosol optical depth and solar zenith angle dominate; the azimuth and view-zenith coordinates are almost inert at first order.

approximately 0.1 for elevation and water vapor. The two remaining angles have smaller effects (Figure 2, right). These validation diagnostics motivate input-specific kernel scales. They measure sensitivity of the fitted linear model, rather than identifying an intrinsic dimension of the regression problem.

Low-degree surrogates are accurate. Linear regression on the standardized state achieves $R^2 = 0.83$, compared with 0.74 for distance-weighted five-nearest-neighbors. Cubic polynomial ridge regression has mean relative component error of approximately two percent (Table 1). Together with the spectral compression, these results motivate smooth low-dimensional regression as a baseline.

Reflectance inversion requires additional care. In deep absorption windows, the transmitted flux $Y_2 + Y_3$ can be as small as 10^{-21} . At a fully opaque band, the radiance is independent of surface reflectance and the inverse is non-identifiable. An auxiliary round-trip calculation on 2,332 states gives reflectance RMSE 0.0054 even with exact tabulated components. Its error is concentrated in 1,652 of 664,620 sample–band entries, whose inverse denominators are below 2×10^{-13} . All returned inverses are finite. This is a floating-point conditioning diagnostic, not a lower bound on emulator error. We report both the center and the tails of the reflectance-error distribution. Non-finite inverses are assigned zero by the scoring routine; their frequency is recorded separately.

Differentiating (2) separates the components. Write $D = \hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)$ for its denominator. Then

$$\frac{\partial \hat{\rho}}{\partial \hat{Y}_4} = -\hat{\rho}^2, \quad \frac{\partial \hat{\rho}}{\partial \hat{Y}_2} = \frac{\partial \hat{\rho}}{\partial \hat{Y}_3} = -\frac{\hat{\rho}}{D}, \quad \frac{\partial \hat{\rho}}{\partial \hat{Y}_1} = -\frac{\hat{Y}_2 + \hat{Y}_3}{D^2}. \quad (3)$$

At exact identifiable components, $\hat{\rho} = \rho = 0.7$, so the albedo derivative is -0.49 , independently of the positive transmitted flux. For general predictions it equals $-\hat{\rho}^2$ and can be large. The denominator factors in (3) make path-radiance and flux errors particularly important near opaque bands. Section 5.1 reports the resulting all-band error tails.

3.1 Finite-perturbation control of the inversion

The derivatives in (3) describe local conditioning. The following elementary identities give a finite-error test and distinguish forward accuracy from inversion stability. For one sample and band, write $a = Y_1$, $t = Y_2 + Y_3$, $s = Y_4$ and $L = a + \rho t / (1 - \rho s)$. Set $e_a = \hat{a} - a$, $e_t = \hat{t} - t$, $e_s = \hat{s} - s$.

Proposition 1 (Conditional radiance and reflectance stability). *Suppose $0 \leq \rho \leq \rho_{\max}$, $t > 0$ and $q = 1 - \rho s > 0$. If also $\hat{q} = 1 - \rho \hat{s} > 0$, the forward error satisfies the exact identity*

$$\hat{L} - L = e_a + \frac{\rho e_t}{\hat{q}} + \frac{\rho^2 t e_s}{q \hat{q}}. \quad (4)$$

In particular, if $q, \hat{q} \geq \kappa > 0$,

$$|\hat{L} - L| \leq |e_a| + \rho_{\max} |e_t| / \kappa + \rho_{\max}^2 |t e_s| / \kappa^2.$$

For inversion at the true radiance, put $u = L - a = \rho t / q$, $D = t / q$, and $\hat{D} = \hat{t} + \hat{s}(L - \hat{a})$. Whenever $\hat{D} \neq 0$,

$$\hat{\rho} - \rho = -\frac{(1 - \rho \hat{s})e_a + \rho e_t + \rho u e_s}{\hat{D}}, \quad \hat{D} - D = e_t + u e_s - \hat{s} e_a. \quad (5)$$

If $B = |e_t| + |u e_s| + |\hat{s} e_a| \leq \eta D$ for some $0 \leq \eta < 1$, then $\hat{D} \geq (1 - \eta)D > 0$ and

$$|\hat{\rho} - \rho| \leq \frac{|1 - \rho \hat{s}| |e_a| + \rho |e_t| + \rho |u e_s|}{(1 - \eta)D}. \quad (6)$$

Proof in Appendix A. When $\hat{q} \neq 0$, the same calculation gives the exact forward-inverse link

$$\hat{\rho} - \rho = -\frac{\hat{q}}{\hat{D}}(\hat{L} - L). \quad (7)$$

Consequently, on a common set where $|\hat{q}| \leq Q$ and $|\hat{D}| \geq \delta > 0$, the corresponding unweighted L^2 errors obey $\|\hat{\rho} - \rho\|_2 \leq (Q/\delta)\|\hat{L} - L\|_2$. Without the denominator condition there is no uniform implication from small spectral relative radiance error to small reflectance error. For example, take two bands with $a = s = 0$, $t = (1, \varepsilon)$ and $0 < \rho < 1$. A predictor with $\hat{t} = t$, $\hat{s} = 0$ and $\hat{a} = (0, c\varepsilon)$, where $0 < c < \rho$, has relative spectral radiance error $c\varepsilon/(\rho\sqrt{1 + \varepsilon^2}) \rightarrow 0$ but retrieves $(\rho, \rho - c)$, so its reflectance RMSE remains $c/\sqrt{2}$. Every denominator is positive for $\varepsilon > 0$ and both retrieved reflectances remain physical. This example isolates the loss of inversion stability as the second-band flux vanishes.

The forward bound need not deteriorate when t tends to zero, whereas the inverse bound requires error control relative to $D = t/q$. Thus a small spectral radiance error does not certify stable reflectance recovery. At $t = 0$, $L = a$ is independent of ρ wherever the forward formula is defined, so reflectance is unidentifiable from this band. The identities are exact-arithmetic statements; floating-point cancellation in $L - a$ can further affect an implemented inverse.

For empirical comparison, a common application passband and conditioning threshold should be fixed before model evaluation. The utility `code/conditioned_reflectance.py` uses a positive flux scale determined on training data and a common truth-based mask, which is available in simulation benchmarks. It reports retained coverage, non-finite inversions and denominator failures separately. Masked errors require sample-level arrays and cannot be recovered from aggregate summaries. The EMIT comparisons in Section 5.1 therefore use the all-band statistics.

4 Models

All models are trained per component and evaluated in physical units after undoing the standardization; unless stated otherwise they regress the 64 leading principal coefficients of the standardized component, with the training variance criterion and projection qualification of Section 3. Every hyperparameter named below is selected on the validation set of the seed at hand. The comparisons of Sections 5–6.1 run on CPU; the ten-kernel comparison, the regularisation suite and the matched benchmark of Sections 10–12 run on one T4 GPU, whose device and peak memory each run record carries.

Reference family. Cubic polynomial ridge regression [43], acting on all monomials of degree at most three in the six standardized inputs, with the ridge weight selected from $\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. It provides a low-cost, low-degree reference; its empirical error alone does not prove a smoothness class for the underlying function.

Exact kernel ridge regression. Three constructions, all Matérn kernels [45] on the standardized inputs with the length scale expressed as a multiple of the median pairwise distance of the fit set. The listed nugget is γ in $(K + n\gamma I)\alpha = Z$, not the unscaled diagonal term. The subset baseline is a Matérn-5/2 kernel fitted on a uniformly drawn subset of 4,000 training points — the landmark choice of a Nyström approximation [60, 61] — one eigendecomposition per length scale, multiplier in $\{1, 2\}$, nugget in $\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. The second is the exact solve on every training row: smoothness $\nu \in \{3/2, 5/2\}$, multiplier in $\{0.5, 1, 2, 4\}$ and the same nugget grid, tuned on validation from a 6,000-row training subsample and the winner refit on all 18,884 rows by one Cholesky factorization [107]. The third gives each input its own length scale, an automatic relevance determination [90, 8]: starting from the isotropic winner, a coordinate search over per-input multipliers in $\{1/4, 1/2, 1, 2, 4\}$, two sweeps, scored on validation, after which the smoothness, scale and nugget are re-tuned at the found metric and the solve is refit on every row. The third construction allows the kernel to adapt to the input anisotropy measured in Section 3.

Fully connected networks. Three ELU hidden layers of 512 units per component, trained on the 285 standardized bands with AdamW [86, 87] at learning rate 10^{-3} , cosine decay, batch size 1024, weight decay 10^{-6} , up to 150 epochs with early stopping on validation loss [88] (patience 25). Five networks are trained per component and seed from different initializations; the table reports the first and the mean of the five.

Kernel corrections of the neural mean. The construction of [12], with the exact kernel stages fitted on every training row. (i) *Residual correction*: an exact Matérn regression, on the standardized input space, of the network’s training residuals in coefficient space, added back to the network’s projection into that space, with smoothness, scale and nugget selected by the validation error of the corrected prediction; applied to the single network and to the five-network mean. (ii) *Kernel on features*: the same exact regression with the input metric replaced by the network’s last hidden layer, standardized, giving a kernel metric in the fitted feature representation. This is the post-hoc counterpart of deep kernel learning [10], in which the representation and the kernel are trained together, and of the kernel-flow regularization of a network’s inner layers [11]; here the representation is fixed first and the kernel is solved exactly on it. (iii) *Per-coordinate selection*: for each of the 64 output coefficients independently, pick on

model	component rel. L^2 [%]				radiance	reflectance
	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [pp]
Cubic ridge	1.16±0.02	2.18±0.04	1.11±0.03	2.98±0.06	0.760±0.011	0.471±0.010
Matérn KRR, 4000-point fit	0.17±0.01	0.61±0.04	0.29±0.02	2.44±0.87	0.332±0.029	0.178±0.030
Matérn KRR, exact on all rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
Matérn KRR, per-input scales	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
FC-DNN (3×512)	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
Five-network mean	0.37±0.01	0.67±0.03	0.28±0.01	2.51±0.09	0.251±0.006	0.163±0.003
DNN + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
Ensemble + residual KRR	0.08±0.00	0.27±0.03	0.13±0.00	2.42±0.17	0.123±0.007	0.080±0.008
Kernel on network features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
Per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
Convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003

Table 1: Test errors under the protocol of Section 3, mean $\pm$ sample standard deviation over ten splits. Component and radiance columns report mean relative L^2 error; radiance is evaluated at $\rho = 0.7$. The last column is median absolute reflectance error in percentage points. The inverse is evaluated across all bands, including deep absorption windows, with the non-finite convention of Section 3. Table 3 reports the associated 95th-percentile errors and RMSE.

validation among the exact kernel, the network, the ensemble, the two corrected predictors and the feature kernel, and assemble the test prediction from the winners. (iv) *Convex stack*: per component, a convex combination of the same six heads with weights fitted on the validation points in the physical relative metric, in the manner of stacked generalization [80] and stacked regression [81]. The choice of weighting rule for pools of this kind, and its relation to constrained minimum-variance weighting, is studied separately and is not part of this comparison.

Sample-size, rank and width experiments. Section 6 repeats the isotropic exact kernel, the network, its residual correction and the feature kernel at nested training sizes, at other output ranks and at a wider network; the table of Section 5 is the full-size, rank-64, width-512 cell of that grid.

5 Results

Table 1 compares the model families at 18,884 training rows, output rank 64 and network width 512.

Polynomial and subset-kernel baselines. Cubic ridge regression gives 1.86% mean component error and 0.760% radiance error. The 4,000-point Matérn fit reduces these means to 0.88% and 0.332%. Its albedo error varies substantially across splits: approximately 1.9% on seven splits and 3.3–4.3% on the remaining three.

Full-data and input-scaled kernels. The isotropic full-data kernel reduces mean radiance error to 0.239% and path-radiance error to 0.08%, compared with 0.17% for the subset fit. Its albedo error is 2.3–2.9% across splits: less variable, but larger than the subset fit on the latter’s seven most accurate splits. Input-specific length scales give 0.095% radiance error, 0.63% mean component error and 0.036 percentage points of median reflectance error. The fitted scales are consistent with weak azimuth dependence of the flux components. The relative-azimuth length

scale is sixteen to thirty-two times larger than the other scales, with median factors sixteen for direct flux and albedo and thirty-two for diffuse flux. The longer azimuth scale occurs on every split for the flux components and nine of ten for albedo. View-zenith scales are two to sixteen times longer, whereas the path-radiance metric is approximately isotropic.

Networks and kernel corrections. The network has 1.22% mean component error and 0.387% radiance error. Averaging five initializations reduces these to 0.96% and 0.251%. Residual kernel regression reduces the single-network radiance error to 0.141% and the ensemble error to 0.123%. The feature kernel has 0.61% mean component error and 0.098% radiance error. Its radiance comparison with the input-scaled kernel is close: the feature kernel is lower on six of ten splits, although its mean is slightly larger. Its median reflectance error is 0.054 points, compared with 0.036 for the input-scaled kernel, and is larger on every split. The separately tuned residual correction substantially improves the network, but has larger radiance error than the input-scaled and feature kernels.

Selection and stacking. Both combiners use the isotropic full-data kernel, the network, its five-member ensemble, the two residual-corrected predictors and the feature kernel. The input-scaled kernel is not in this pool. Coordinatewise validation selection gives 0.193% radiance error, larger than several individual members. The convex stack gives 0.087% radiance error, 0.57% mean component error and 0.048 points of median reflectance error. Its radiance gain is approximately 8% relative to the input-scaled kernel and 11% relative to its best individual member, the feature kernel. On mean component error, the stack improves on the input-scaled kernel at nine splits, with paired difference 0.06 ± 0.05 points, and on the feature kernel at all ten, with difference 0.047 ± 0.006 points.

The residual second moments provide a diagnostic of complementarity. On path radiance, the isotropic kernel and five-network ensemble have residual correlation 0.27 ± 0.09 , below their error ratio of approximately 0.34 at every split. For a quadratic error metric, this is the classical condition under which a convex combination can improve on the better member [79, 12]. The stack instead minimizes a mean relative norm, and its reflectance objective is nonlinear; the diagnostic does not guarantee improvement in those metrics. The two residual-corrected predictors have much larger correlations, approximately 0.88 on path radiance and 0.99 on albedo. Coordinatewise selection also differs from stacking in its objective and number of independently chosen coefficients.

Spherical albedo. Albedo has the largest relative error for every family. Its small short-wave-infrared values make relative errors sensitive to absolute prediction errors in that region. The width-512 stack achieves 1.94% albedo error. The learning curves below show little improvement with sample size on this component; the experiments do not isolate whether the plateau arises from approximation, optimization or the relative-error weighting.

5.1 Paired comparisons and the radiance–reflectance trade-off

Paired differences compare candidate and reference errors on the same split. Table 2 reports those differences for the main and wide pipelines. Positive gains favor the candidate. The wide feature kernel improves on the input-scaled kernel at all ten splits, by 0.0165 ± 0.0066 radiance percentage points. By contrast, the wide stack’s improvement over its own feature kernel is only 0.0006 ± 0.0026 points, with eight wins and two losses. The mean improvement is small relative

candidate / reference	mean gain $\pm$ SD [pp]	gain range [pp]	wins/10
Input-scaled / isotropic KRR	0.1436 $\pm$ 0.0122	[0.1196, 0.1585]	10
Residual correction / network	0.2464 $\pm$ 0.0090	[0.2365, 0.2615]	10
Narrow stack / input-scaled KRR	0.0083 $\pm$ 0.0066	[-0.0054, 0.0185]	9
Wide feature kernel / input-scaled KRR	0.0165 $\pm$ 0.0066	[0.0054, 0.0295]	10
Wide / narrow feature kernel	0.0191 $\pm$ 0.0079	[0.0096, 0.0352]	10
Wide stack / wide feature kernel	0.0006 $\pm$ 0.0026	[-0.0053, 0.0037]	8

Table 2: Paired radiance contrasts on seeds 101–110. Gain is the reference error minus the candidate error, in percentage points (pp). SD is the sample standard deviation of the ten paired differences; the range and win count describe those same ten differences. They are not confidence intervals or tests of a population-level superiority claim.

model	radiance [%]	median [pp]	95th percentile [pp]	RMSE [pp]
Cubic ridge	0.760 $\pm$ 0.011	0.471 $\pm$ 0.010	70.68 $\pm$ 0.02	4.18e+03
Isotropic KRR	0.239 $\pm$ 0.016	0.141 $\pm$ 0.018	69.53 $\pm$ 0.42	4.18e+03
Input-scaled KRR	0.095 $\pm$ 0.009	0.036 $\pm$ 0.004	31.16 $\pm$ 6.21	1.34e+04
Narrow feature kernel	0.098 $\pm$ 0.007	0.054 $\pm$ 0.004	7.75 $\pm$ 1.13	1.78e+03
Narrow convex stack	0.087 $\pm$ 0.006	0.048 $\pm$ 0.003	29.44 $\pm$ 5.89	1.25e+03
Wide feature kernel	0.079 $\pm$ 0.009	0.045 $\pm$ 0.006	7.57 $\pm$ 0.95	1.40e+03
Wide convex stack	0.078 $\pm$ 0.011	0.043 $\pm$ 0.005	22.71 $\pm$ 4.42	5.81e+03

Table 3: All-band metrics from the complete ten-seed main and wide campaign records. Radiance is mean relative L^2 error. Reflectance median, 95th-percentile absolute error and RMSE are in percentage points at $\rho = 0.7$. The first three columns show mean $\pm$ sample SD across splits; RMSE shows its mean only, with its spread retained in the machine-readable summary. The inverse is neither physically clipped nor conditioned on an application passband. All inverses in these records are finite, so their zero-fill branch does not explain the large errors. Large finite values near denominator zeros dominate RMSE. These are diagnostic all-band errors, not operational retrieval accuracy.

to the variability of the paired differences. These are descriptive paired summaries: repeated splits of one finite table, overlapping test rows and adaptive study development do not supply ten independent external validation experiments.

Radiance and reflectance also order the heads differently. Table 3 reports the all-band reflectance tails alongside the medians. Each entry averages a within-split statistic; the mean of ten 95th percentiles is not a pooled 95th percentile. For the width-512 pipeline, the convex stack lowers radiance error relative to the feature kernel, but raises the average 95th-percentile reflectance error from 7.75 to 29.44 percentage points. The tail is worse at all ten splits, by factors between 2.63 and 4.94. The width-2000 comparison repeats the reversal: the feature kernel has 7.57 points at the 95th percentile and the stack 22.71, with the stack worse at all ten splits by factors between 2.36 and 3.79. Thus the near-tie at 0.079% versus 0.078% radiance error conceals a substantial difference in the inverse metric.

The input-scaled kernel illustrates the same distinction: its median error is only 0.0365 points but its mean 95th percentile is 31.16 points. Conversely, the feature kernels have slightly larger

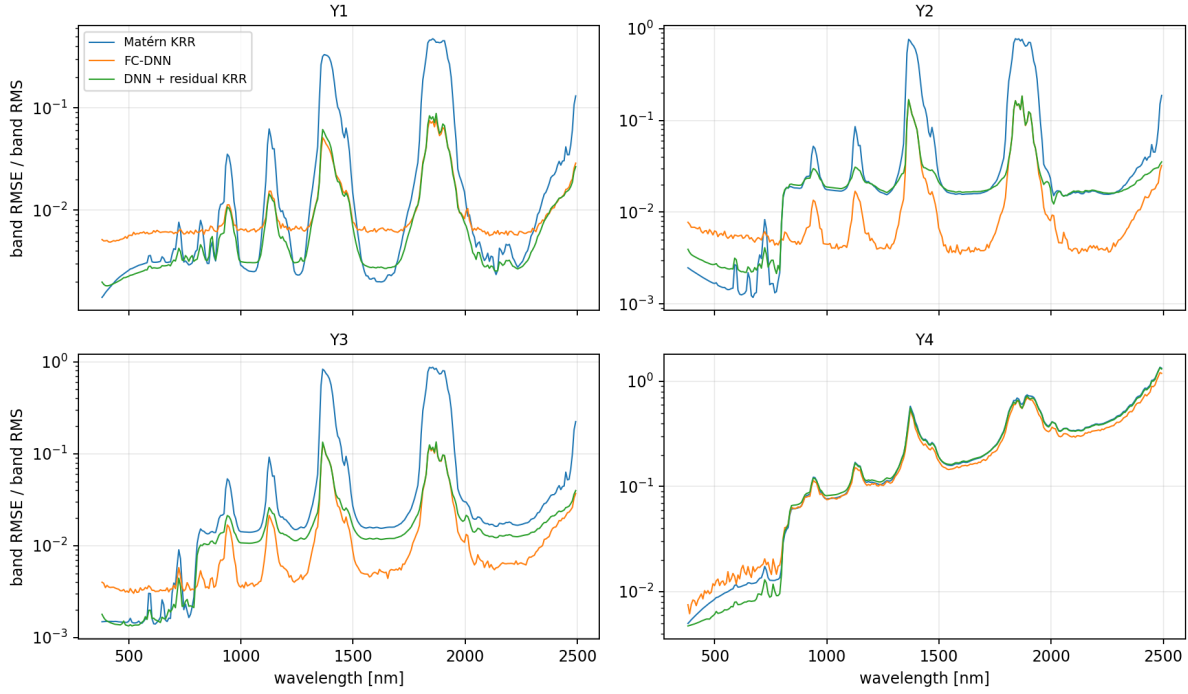


Figure 3: Where along the spectrum each family makes its error: band RMSE divided by the band’s root-mean-square value over the test rows, against wavelength, for the isotropic Matérn kernel, the 3×512 network and the network with its residual kernel. The kernel’s error follows the band structure, rising by two orders of magnitude in the water-vapour bands near 940, 1140, 1380 and 1870 nm; the network’s is nearly flat across the spectrum and higher in the windows; the residual correction keeps the network’s floor in the windows and most of the kernel’s advantage outside them. On the spherical albedo Y_4 the three families agree to within a factor of two at every wavelength, and all of them degrade towards 2500 nm. These bandwise relative-RMSE diagnostics use a different aggregation from the mean per-sample relative spectral norm in Table 1.

medians and appreciably smaller tails. Model rankings depend on the metric and spectral domain. The identities of Section 3.1 relate inverse error to the predicted denominator as well as forward error. This explains why optimizing componentwise or radiance error alone need not control reflectance tails; it does not identify the component-error correlations responsible for a particular observed tail.

The tables are generated from the twenty individual records. The generator checks split configurations, agreement of recorded array digests, and equality of common deterministic baselines. These checks establish consistency of the summaries; reproducing model training requires the raw arrays.

6 Learning curves, output rank and network width

The following experiments vary training size, output rank and network width, using the same validation-based selection. Learning curves include cubic ridge, isotropic and input-scaled kernels,

training rows	seeds	Cubic ridge	KRR, isotropic	KRR, per-input	FC-DNN	DNN + residual KRR	kernel on features
500	10	0.846±0.031	0.639±0.022	0.415±0.024	1.961±0.057	0.815±0.035	0.456±0.020
1,000	10	0.802±0.024	0.519±0.034	0.260±0.066	1.862±0.015	0.590±0.027	0.316±0.040
2,000	10	0.780±0.016	0.424±0.051	0.217±0.052	1.232±0.034	0.376±0.041	0.240±0.049
4,000	10	0.769±0.010	0.328±0.028	0.154±0.037	0.830±0.019	0.256±0.021	0.164±0.033
8,000	10	0.763±0.010	0.279±0.020	0.119±0.027	0.584±0.020	0.190±0.014	0.121±0.017
18,884	10	0.760±0.011	0.239±0.016	0.095±0.009	0.387±0.012	0.141±0.006	0.098±0.007

Table 4: Learning curves: radiance relative L^2 error [%] against the number of training rows, mean $\pm$ standard deviation over the ten splits reported in the archived aggregate. At $n = 8,000$, one individual record per kernel campaign is absent from the public archive; the table reports the ten-run aggregate. Each rung uses the first n rows of the seed’s training block; the validation and test blocks are those of Table 1, so the top rung is that table. The per-input kernel repeats its coordinate search at every rung. Component columns are in Figure 4 and in the per-seed records.

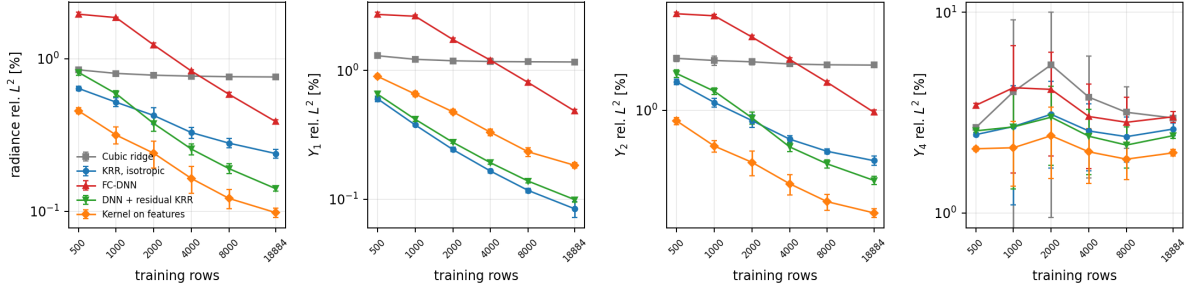


Figure 4: Learning curves on log–log axes for the radiance and for three of the four components, mean and standard deviation over ten splits. Cubic error changes little over the sample-size range. The kernel, network, residual-corrected network and feature kernel decrease approximately as $n^{-1/2}$ on the first three components, with fitted slopes in Table 5. Albedo error shows no sustained decrease; unusually large errors on individual splits increase its spread at 1,000 and 2,000 rows.

the network, residual correction and the feature kernel. Ensemble comparisons use the full training size. The number of available repetitions is given with each experiment.

Learning curves. Each rung uses the first n rows of the seed’s training block, $n = 500, 1,000, 2,000, 4,000, 8,000$ and the full 18,884, with the validation and test blocks unchanged, so the rungs of one seed are nested and the full block is the top rung (Table 4, Figure 4). Cubic error changes little: from 0.85% to 0.76% over a factor of 38 in data, with most of its decrease achieved by a thousand rows. On the first three components every other family decays at close to $n^{-1/2}$ (Table 5): the least-squares slopes of log error against log n over the six rungs lie between -0.38 and -0.61 for the kernel, the network, the corrected network and the feature kernel on Y_1 , Y_2 and Y_3 , while on the albedo every slope lies between -0.09 and $+0.01$: the albedo shows no sustained decrease over this budget range. This plateau does not distinguish approximation bias, optimization, relative-error weighting or limitations of the training data.

These are slopes of six mean errors over a finite sample-size range, not established asymptotic rates. Standard Sobolev or kernel regression rate formulas require sampling, source and regularization assumptions [46, 48]; neither those assumptions nor the smoothness norm of the

family	slope over the six rungs					slope of the last step				
	Y_1	Y_2	Y_3	Y_4	radiance	Y_1	Y_2	Y_3	Y_4	radiance
Cubic ridge	-0.03	-0.03	-0.03	-0.02	-0.03	-0.00	-0.01	-0.00	-0.08	-0.01
KRR, isotropic	-0.55	-0.38	-0.40	-0.01	-0.28	-0.38	-0.19	-0.17	0.10	-0.18
KRR, per-input	-0.45	-0.43	-0.43	0.01	-0.40	-0.32	-0.26	-0.29	0.12	-0.26
FC-DNN	-0.50	-0.49	-0.57	-0.09	-0.48	-0.59	-0.60	-0.55	0.07	-0.48
DNN + residual KRR	-0.52	-0.53	-0.61	-0.04	-0.50	-0.39	-0.33	-0.39	0.12	-0.35
Kernel on features	-0.45	-0.44	-0.46	-0.03	-0.43	-0.28	-0.22	-0.31	0.08	-0.25

Table 5: Learning-curve exponents: the least-squares slope of log error against $\log n$ over the six rungs of Table 4, and the slope of the last step alone, from 8,000 to 18,884 rows, per family and component.

fitted network residual is verified by this experiment. The isotropic kernel’s decrease is slower at the largest sample size. Its slope over the first five rungs is -0.59 , -0.44 and -0.47 on the smooth components, and its last step, from 8,000 to 18,884 rows, is -0.38 , -0.19 and -0.17 , while the network’s last step is as steep as its average (-0.59 , -0.60 and -0.55 ; Table 5). Part of this is the tuning protocol: the kernel’s smoothness, scale and nugget are chosen on a 6,000-row subsample at every rung, so above 6,000 rows the hyperparameters are those of a smaller problem, and Section 6.1 measures what tuning with every row in the solve recovers: a substantial part of the measured change in slope. The consequence for the comparison is that the ratio of the network’s radiance error to the kernel’s falls from 3.1 at 500 rows to 1.6 at the full block, or to 1.8 with the kernel tuned on every row. The kernel’s margin over the network on this table is a margin at this training size; whether it would survive ten times the data depends on where the kernel’s curve settles, which the bend leaves open.

Relative to the isotropic kernel, the per-input metric changes the measured finite-range slope as well as the error level. Its coordinate search, repeated at every rung, gives a slope of -0.399 on the radiance against the isotropic kernel’s -0.279 , and the ratio between them falls from 0.65 at 500 rows to 0.40 at the full block, with one step against that direction (0.50 at 1,000 rows, 0.51 at 2,000) and a fall at every other. The relative error reduction increases over this sample-size range. The selected metric, smoothness and nugget all change with sample size, so these measurements do not isolate a change of asymptotic dimension or establish a faster universal rate. Against the kernel on network features the margin closes as the data grows. The per-input metric has the lower mean at all six rungs, but the split-by-split count runs 10, 9, 10, 7, 5 and 6 wins of ten from 500 rows to the full block, and the gap itself narrows from 0.415 against 0.456 at 500 rows to 0.095 against 0.098 at the top; the feature kernel’s own slope, -0.435 , is the steeper of the two. The input-scaled kernel has a larger advantage at smaller sample sizes, while the feature kernel approaches it as the training set grows. On the albedo the per-input slope is $+0.01$: with no sustained improvement over this range.

The corrected network is below both of its parents on the radiance at every one of the ten splits from 2,000 rows on, and at no split at 500 or 1,000 rows, where its error sits between the network’s and the kernel’s. On Y_2 and Y_3 it is below both at every split at the two largest sizes; on Y_1 it never is, the isotropic kernel alone being the better head at every size (0.084% against 0.099% at the full block). Its exponent is close to the network’s and its error level is closer to the kernel’s in these runs. Section 7 gives the exact residual decomposition, while distinguishing it from a proof of these empirical slopes or of the component-specific exception.

PCA rank	variance outside the basis	KRR, isotropic	FC-DNN	DNN + residual KRR	kernel on features
16	7.5×10^{-4}	0.270±0.015	0.390±0.015	0.175±0.007	0.148±0.009
32	1.5×10^{-5}	0.243±0.018	0.390±0.015	0.139±0.006	0.102±0.009
64	8.2×10^{-8}	0.242±0.018	0.390±0.015	0.143±0.006	0.100±0.009
128	5.4×10^{-9}	0.242±0.018	0.390±0.015	0.150±0.005	0.100±0.009
285	0	0.242±0.018	0.390±0.015	0.172±0.004	0.100±0.009

Table 6: Output-rank ablation: radiance relative L^2 error [%] when the kernels and corrections regress the leading r principal coefficients, mean $\pm$ standard deviation over five splits. The second column is the fraction of the standardized variance the basis leaves out for the slowest-decaying component, the spherical albedo; at rank 285 it is zero. The network regresses the 285 bands directly and is the same predictor in every row.

network	head	component rel. L^2 [%]				radiance rel. L^2 [%]	reflectance med. $ \hat{\rho} - \rho $ [pp]
		Y_1	Y_2	Y_3	Y_4		
3×512	network	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
	network + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
	kernel on its features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
	per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
	convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003
3×2000	network	0.93±0.20	2.11±0.49	0.76±0.19	3.27±0.19	0.748±0.116	0.508±0.110
	network + residual KRR	0.12±0.01	0.35±0.03	0.19±0.03	2.35±0.13	0.160±0.016	0.106±0.010
	kernel on its features	0.13±0.02	0.13±0.02	0.08±0.01	2.10±0.34	0.079±0.009	0.045±0.006
	per-coordinate selection	0.14±0.02	0.30±0.26	0.09±0.01	2.55±0.26	0.187±0.057	0.099±0.033
	convex stack	0.08±0.01	0.13±0.02	0.08±0.00	2.00±0.24	0.078±0.011	0.043±0.005

Table 7: The pipeline of Section 4 run with a 3×2000 network in place of the 3×512 one, at the same ten splits: the network (one member, up to 500 epochs under the same optimizer, learning rate and early stopping), its residual correction, the exact kernel on its last-layer features, and the per-coordinate selection and convex stack over the wide pipeline’s heads (the isotropic kernel, the network, its correction and its feature kernel), beside the corresponding rows of Table 1; mean $\pm$ standard deviation over the ten splits.

Output rank. The kernels and corrections regress the leading r principal coefficients of the standardized component; the network regresses the 285 bands and is the same predictor at every rank. Table 6 refits at $r = 16, 32, 128$ and 285 at five of the seeds beside the rank-64 rows of those seeds. The kernel’s measured error changes little above rank 32: its radiance error reads 0.243, 0.242, 0.242 and 0.242 at 32, 64, 128 and 285, and 0.270 at 16, where the truncation leaves 7.5×10^{-4} of the albedo’s variance outside the basis. The kernel on features behaves the same way. The residual correction does not: it is best at 32 (0.139), a little worse at 64 (0.143), and worse again at 128 and 285 (0.150 and 0.172). The two regressions fit different targets: the direct kernel fits the component, whereas the correction fits the projected network residual. Increasing rank therefore exposes the correction to additional network-error directions, even when the target variance in those directions is small. The lowest measured correction errors occur at ranks 32–64. The projection identity in Section 7 separates these two error terms.

A wider network. A network of three hidden layers of 2,000 units, trained for up to 500 epochs under the same optimizer, learning rate and early stopping as the 512-unit network, was run at all ten seeds with the rest of the pipeline built on it (Table 7). It is a worse emulator at

every seed, 0.75% against 0.39% on the radiance, and its early stopping fired between 42 and 210 epochs of the 500 allowed: at this learning rate and batch size this particular wider-network training configuration gives worse standalone predictions. Width and epoch budget change together; these results do not rule out a better-optimized wide network. The exact kernel on its last-layer features reaches 0.079% on the radiance against 0.098% on the 512-unit features, below the per-input kernel’s 0.095% at all ten seeds and below the convex stack of Table 1 at nine of them, and on the two flux components, at 0.13% and 0.08%, it is at or below every row of that table. The convex stack over the wide pipeline’s heads, which puts 0.88 to 0.96 of its weight on that feature kernel, has the lowest mean radiance error in the ten-split comparison at 0.078% on the radiance and 0.043 reflectance points, below the stack of Table 1 at nine of the ten seeds; the per-coordinate selection over the same heads does not transfer here either. The residual correction of the wider network is a little worse than that of the narrower one, although that ordering is not guaranteed merely because its base network is worse (Section 7). A 3×1000 rung at the same ten seeds sits between the two: the network at 0.72%, the kernel on its features at 0.088% against 0.098% at width 512 and 0.079% at width 2,000. From 512 to 2,000 the network’s own error rises with width while the error of a kernel on its features falls: the fitted features improve kernel prediction despite the larger standalone network error. At 4,000 units, run at two seeds, the network is worse again at 0.95% and the kernel on its features turns back up to 0.097%, so the gain in the representation does not continue past 2,000.

A three-member wide-network experiment, run at three seeds, supplies the exact kernel with the concatenated last-layer features of all three members at once, 6,000 coordinates rather than 2,000. It reads 0.074% on the radiance against 0.083% for the kernel on a single member’s features on the same splits, lower at all three seeds, and 0.043 against 0.051 reflectance points, again at all three. That puts one exact solve level on the radiance with the convex stack of six heads fitted on those same splits (0.074% against 0.074% on the mean, though the stack is the lower at two of the three seeds and on the reflectance median at all three). The six-predictor stack and the single kernel on concatenated features have similar mean radiance errors on these three splits. The comparison does not isolate all optimization and kernel-selection effects. Three seeds provide exploratory evidence rather than a ten-split replication, and the concatenated-feature construction was evaluated only at width 2,000.

6.1 Full-data tuning and gradient-based metrics

We compare hyperparameter selection on a 6,000-row subsample with selection on all training rows, and compare coordinate search with a metric derived from a cubic surrogate’s gradients (Table 8). The reported aggregates contain ten splits; the per-split archive for full-data tuning contains five of them.

Tuning with every row in the solve. The kernel families of Section 4 choose smoothness, scale and nugget by the validation error of a solve on a 6,000-row subsample, and refit the winner on all 18,884 rows. Doing the tuning solve on every row instead lowers the isotropic kernel’s radiance error from 0.239% to 0.210% and its reflectance median from 0.141 to 0.107 points, better at nine of the ten splits, the ratio averaging 0.882 and never above 1.001. The selected scales differ: over the forty component-seed cells the full block picks a shorter length scale than the subsample in 26, the same in 12, and a longer one in 2. In this experiment, subsample selection usually chooses a longer length scale than full-data selection, consistent with the smaller tuning fit favoring greater smoothing.

kernel	component rel. L^2 [%]				radiance	reflectance
	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [%]
isotropic, tuned on 6,000 rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
isotropic, tuned on every row	0.08±0.00	0.36±0.02	0.17±0.01	2.58±0.14	0.210±0.008	0.107±0.006
per-input search, tuned on 6,000 rows	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
per-input search, re-tuned on every row	0.08±0.00	0.16±0.02	0.08±0.01	2.09±0.12	0.096±0.009	0.036±0.004
per-input from the cubic’s gradients, $p = 1$	0.34±0.01	0.19±0.02	0.14±0.01	2.17±0.14	0.159±0.007	0.078±0.004
per-input from the cubic’s gradients, $p = 1/2$	0.11±0.00	0.19±0.02	0.09±0.01	2.27±0.15	0.125±0.008	0.055±0.004
per-input from the cubic’s gradients, $p = 1/4$	0.08±0.00	0.25±0.02	0.14±0.01	2.38±0.20	0.157±0.015	0.077±0.013

Table 8: Where the kernel’s hyperparameters are chosen, and how its metric is obtained, at the full training block and the same ten splits. The first pair tunes smoothness, scale and nugget on a 6,000-row subsample and on every training row; the last three rows replace the coordinate search by a metric read off the cubic surrogate’s gradients, at three exponents. Selecting the exponent on validation, per seed, chooses $p = 1/2$ at all ten, so that row would duplicate the $p = 1/2$ row and is not printed; the per-seed choices are in the stored numbers. Mean $\pm$ standard deviation over the ten splits.

Below 6,000 rows the tuning subsample *is* the whole training block and the rungs are already full-block tuned; at 8,000 rows it is three quarters of it; only the top rung is a third. Applying the measured ratio to the top rung of Table 4 takes the kernel’s last step on the radiance from -0.183 to -0.334 , against a slope of -0.305 over the five lower rungs: the top-step slope becomes closer to that of the lower rungs. One retuned endpoint does not establish a power law over or beyond the observed range. On the two flux components the same correction takes the last step from -0.187 and -0.171 to -0.352 and -0.358 against averages of -0.437 and -0.468 , reducing, but not eliminating, the difference between the last-step and lower-rung slopes. The consequence for the comparison of Section 5 is that the kernel’s margin over the network at the full block is wider than Table 4 shows: 1.8 times rather than 1.6, under this retuned endpoint. No conclusion about unobserved sample sizes follows.

Full-data tuning has little effect on the input-scaled kernel: 0.096% against 0.095%, better at four of the ten splits, ratio 1.005. Its coordinate search already selects a scale of once the median distance at the metric it finds, which is what the full-block tune selects too. The input-scaled kernel is therefore less sensitive to the tuning sample size in these experiments.

Gradient-based metric. The coordinate search costs a sweep over the six inputs at five multipliers, twice, per component and per seed. The alternative sets the metric before any kernel is fitted: the multiplier of coordinate j is the root-mean-square gradient of the fitted cubic surrogate along j over the training rows, raised to a power p , normalized to geometric mean one and clipped to the search’s own range $[1/16, 4]$, so weakly varying coordinates receive longer length scales. The exponent p controls the strength of the rescaling. It costs one cubic fit and six finite differences.

At $p = 1$, radiance error is 0.159%, below the isotropic kernel at every one of the ten splits by a factor 0.669 on average and above the searched metric at every one of them by a factor 1.680. The dependence on p is nonmonotone: $p = 1/2$ gives 0.125% and $p = 1/4$ gives 0.157%, so the best exponent is interior, and choosing it the way every other hyperparameter in this study is chosen — the lowest mean validation error over the four components, per seed, with the test block untouched — selects $p = 1/2$ at all ten seeds. At that exponent the metric closes 79% of the distance from the isotropic kernel to the searched one on the radiance and 82% on the

reflectance median, for one cubic fit, six finite differences and a choice among three numbers.

The componentwise errors favor different exponents: Y_2 and Y_3 are best at $1/2$, the albedo at 1 (2.18% against 2.27%), and the path radiance at $1/4$. On that component the isotropic kernel already reads 0.084% and the searched metric 0.085%: coordinate search gives little improvement. The proportional metric nonetheless reads 0.34% there, approximately four times the isotropic error. A possible explanation is that the surrogate’s gradient along a coordinate measures how fast the map moves with it and not how much predictive information it carries, so gradient magnitude alone need not provide an appropriate kernel metric. Softening the exponent to $1/4$ takes Y_1 back to 0.085%, exactly the searched metric’s value, at the cost of Y_2 and Y_3 . The validated gradient metric recovers much of the observed radiance gain with fewer metric searches. Coordinate search additionally allows independent componentwise adjustments, whereas the gradient construction selects one exponent for all four components.

7 Residual correction and error alignment

Fix the training design, neural mean and selected kernel hyperparameters. The following identities hold without sampling assumptions. We first work with one output coordinate; the vector-valued statements follow coordinatewise.

7.1 A frozen regression operator

Fix a design $X_n = (x_1, \dots, x_n)$, a positive-semidefinite kernel with Gram matrix K , and $\lambda > 0$. Define

$$(Sg)(x) = k(x, X_n)(K + \lambda I)^{-1}g(X_n), \quad R = I - S. \quad (8)$$

Here λ is the diagonal regularizer in the linear system. The campaign code uses $\lambda = n\gamma$, where γ is its reported nugget; the distinction matters when the training size changes. Although a selected kernel can depend on the targets, S is linear in the response only *after that selection is frozen*.

Proposition 2 (Frozen residual correction). *Let f be the target and h any fitted mean, and set $\hat{f}_c = h + S(f - h)$. Then:*

$$f - \hat{f}_c = R(f - h), \quad (9)$$

$$(Rg)(X_n) = \lambda(K + \lambda I)^{-1}g(X_n). \quad (10)$$

If $Ku_j = \mu_j u_j$, the design residual in direction u_j is multiplied by $\lambda/(\mu_j + \lambda) \in (0, 1]$. The factor is one when $\mu_j = 0$ and strictly below one when $\mu_j > 0$; this is shrinkage, not a hard spectral projection.

Suppose, in addition, that $\mathcal{S}$ is a normed vector space containing f and h and that the following bound holds uniformly on that space for the realized operator:

$$\|Rg\|_{L^2(P)} \leq C_n \|g\|_{\mathcal{S}} \quad (g \in \mathcal{S}).$$

Then

$$\|f - \hat{f}_c\|_{L^2(P)} \leq C_n \|f - h\|_{\mathcal{S}}. \quad (11)$$

Consequently, $\|f - h\|_{\mathcal{S}} < \|f\|_{\mathcal{S}}$ makes this displayed upper bound smaller than $C_n \|f\|_{\mathcal{S}}$ when $C_n > 0$. It is neither a necessary nor a sufficient condition for the corrected predictor’s actual error to be smaller than that of Sf .

Proof in Appendix A.

Why the bounds do not order the errors. On a two-point design take $K = \text{diag}(1, 99)$ and $\lambda = 1$, so $R = \text{diag}(1/2, 1/100)$ and $C_n = 1/2$ is its Euclidean operator norm. With $f = (1, 0)$ and $f - h = (0, 2)$, the residual norm is twice the target norm, yet $\|R(f - h)\| = 0.02 < 0.50 = \|Rf\|$. Conversely, with $f = (0, 1)$ and $f - h = (1/2, 0)$, the residual norm is smaller, but $\|R(f - h)\| = 0.25 > 0.01 = \|Rf\|$. These examples also show why source-norm accuracy alone does not order corrected prediction errors.

7.2 An exact criterion for improvement

The relevant comparison can be made exactly in any Hilbert error metric. This includes a finite weighted squared error and $L^2(P)$, but not directly the mean of per-sample relative norms or the nonlinear radiance and reflectance metrics in Table 1.

Proposition 3 (Error alignment). *For the same frozen operator S , put $a = Rf$ and $b = Rh$. Then*

$$\|f - \hat{f}_c\|^2 - \|f - Sf\|^2 = \|b\|^2 - 2\langle a, b \rangle. \quad (12)$$

Thus full correction improves on Sf exactly when $2\langle a, b \rangle > \|b\|^2$. More generally, the interpolation $p_t = Sf + tRh$, $0 \leq t \leq 1$, has minimum squared error at

$$t_* = \min\{1, \max\{0, \langle a, b \rangle / \|b\|^2\}\} \quad \text{if } b \neq 0; \quad (13)$$

if $b = 0$, every t is equivalent. If the direct fit uses S_0 and the correction uses S_1 , (12) instead holds with $a = (I - S_0)f$ and $b = (S_1 - S_0)f + (I - S_1)h$.

Proof in Appendix A. The criterion depends on the unknown target error. Replacing its inner products by validation errors gives a validation-optimal interpolation, not a guarantee on new data. The reported experiments use separately tuned direct and residual kernels, so their comparison must use the final, two-operator form with their actual selected operators. Neither the criterion nor the training-design contraction establishes that correction will beat both parents at every sample size.

7.3 What the principal-component implementation computes

The campaign trains a network on all standardized bands but projects its predictions into the retained principal-component basis before correction. Thus its corrected predictor is based on the projected network mean. To state the implementation precisely, let f and h now denote the target and network output after the common fitted standardization and PCA centering. Let Π be the orthogonal projection onto the retained spectral coordinates. In these centered coordinates, the implemented prediction is

$$\hat{f}_{c,\Pi} = \Pi h + S\Pi(f - h).$$

The output mean and diagonal band scaling are restored afterwards. Since the same scalar kernel operator acts on each retained output coordinate,

$$f - \hat{f}_{c,\Pi} = (I - \Pi)f + (I - S)\Pi(f - h). \quad (14)$$

The terms on the right are orthogonal in the standardized spectral Euclidean norm at each query, hence their squared norms add there. They need not be orthogonal in physical units after unequal band rescaling. This decomposition separates truncation error from residual regression and explains why output rank is a genuine part of the model specification. It does not by itself prove the observed optimum near rank 32. Comparisons with the unprojected network include the effect of projection as well as residual fitting.

Conditional rate bound. Kernel learning-rate results impose source, spectral, sampling and regularization conditions [46, 47, 48]. If a uniform residual bound holds with $C_n = O(n^{-\alpha})$ and $\|f - h_n\|_{\mathcal{S}} = O(n^{-\beta})$, then (11) gives $\|f - \hat{f}_c\|_{L^2(P)} = O(n^{-(\alpha+\beta)})$. The source-norm bound and the uniformity required for a data-dependent residual are additional hypotheses. They do not follow from empirical network loss. The experiments measure finite-range error slopes, not these source norms or an asymptotic rate.

The code checks the residual identities, regularizer convention, semidefinite boundary, alignment formula, counterexamples and projected implementation on explicit and randomized finite designs.

8 Retrieval-error transfer and the sensitivity of learned corrections

8.1 Sensitivity of the correction

For a common frozen operator $R = I - S_\lambda$ and the norm of Section 7, the exact criterion $2\langle Rf, Rh \rangle > \|Rh\|^2$ is necessary and sufficient for strict improvement of the corrected predictor over the standalone kernel, since $\|Rf - Rh\|^2 < \|Rf\|^2$ is that inequality rearranged; the comparison of smoothness norms is neither necessary nor sufficient. Write the design values of the targets, of the fitted mean and of the corrected predictor as y , h_X and $\hat{y}$, with $\hat{y} = h_X + S_\lambda(y - h_X)$ and $S_\lambda = K(K + \lambda I)^{-1}$; in the nugget convention of the experiments, $\lambda = n\gamma$. The mean is trained on the same labels, so $h_X = h_X(y)$. In the following Jacobian calculation, K and λ are held fixed; label-dependent feature or kernel selection would add terms involving derivatives of S_λ .

Proposition 4. *1. If h_X is differentiable in y , the Jacobian of this fixed-kernel pipeline is $J_{\hat{y}} = S_\lambda + (I - S_\lambda)J_h$. In particular the correction stage alone does not bound the sensitivity of the pipeline: with $K = I$, $\lambda = 1$ and $h_X(y) = 2y$ one has $\hat{y} = \frac{3}{2}y$ although $\|S_\lambda\|_{\text{op}} = \frac{1}{2}$; and if the mean interpolates the training labels, $h_X(y) = y$, then $\hat{y} = y$ and the correction changes nothing on the design.*

2. With the mean and the kernel frozen, $y \mapsto \hat{y}$ is affine with linear part S_λ , its effective degrees of freedom are $\text{tr } S_\lambda = \sum_j \mu_j / (\mu_j + \lambda)$, and $\|S_\lambda\|_{\text{op}} = \max_j \mu_j / (\mu_j + \lambda) < 1$.

3. Suppose the mean, the kernel and the ridge are fixed independently of the correction noise, conditional on the design, and the correction is fitted on an independent sample with $y = f_X + \varepsilon$, $\mathbb{E}\varepsilon = 0$, $\mathbb{E}[\varepsilon\varepsilon^\top] = \sigma^2 I$. With $r = f_X - h_X$ and r_j its coordinates in the eigenbasis of K ,

$$\frac{1}{n} \mathbb{E}_\varepsilon \|\hat{y} - f_X\|^2 = \frac{1}{n} \sum_j \left(\frac{\lambda}{\mu_j + \lambda} \right)^2 r_j^2 + \frac{\sigma^2}{n} \sum_j \left(\frac{\mu_j}{\mu_j + \lambda} \right)^2.$$

4. At a new input x , with $a_\lambda(x) = (K + \lambda I)^{-1}k(X_n, x)$, the noise variance of the correction is $\sigma^2 \|a_\lambda(x)\|_2^2$.

Proof in Appendix A. Item (iii) is what the ridge-path experiments report: the residual's spectral energy $\sum_j r_j^2$ by eigendirection, and the two sums as functions of λ , together with the out-of-sample factor of item (iv). A ridge chosen from fits that depend on the correction noise is outside the hypothesis of (iii).

8.2 A finite-class selection guarantee

A finite comparison on a genuinely independent selection sample has a simple risk bound [82, 83, 84]. The independence assumption is stronger than reusing the same validation rows for training decisions and combination weights.

Proposition 5. *Let $p_1, \dots, p_M$ be predictors frozen before an independent selection sample of m i.i.d. units, let the loss take values in $[0, B]$, and let $\hat{p}$ minimise the empirical risk on that sample. Then with probability at least $1 - \delta$,*

$$R(\hat{p}) \leq \min_{1 \leq j \leq M} R(p_j) + 2B \sqrt{\frac{\log(2M/\delta)}{2m}}.$$

Proof in Appendix A. The squared constrained-retrieval loss is bounded by R^2 . Thus the proposition can compare a finite set of frozen corrections or stacks, with bands and prescribed reflectances averaged within each independent state. For example, a finite grid of damped corrections $h + \alpha k(x, X_n)(K + \lambda I)^{-1}(y - h_X)$ can include $\alpha = 0$; the standalone kernel must be added as a separate candidate when it is not already in the grid. To apply the result to stacks, one first fits their weights on development rows and then compares a frozen finite collection on an independent state-level sample. The experiments below reuse validation rows for early stopping, kernel tuning and combination; they do not implement this additional independent-selection stage. Proposition 5 therefore specifies a possible protocol, not a probabilistic certificate for the reported stacks or overlapping partitions.

8.3 Transfer of component error into retrieval error

Section 3.1 controls the inverse where the denominator stays away from zero. The theorem below describes a whole evaluation distribution, with entries of vanishing transmission included, with the constant one, and shows that the resulting loss of rate is not an artefact of the argument. Let P be the evaluation distribution over entries (a state, a band, and the surface reflectance when it is varied), in fixed physical units; expectations are conditional on the fitted emulator. Assume

$$0 \leq \rho \leq R \leq 1, \quad t > 0, \quad 0 \leq s \leq S, \quad RS < 1. \quad (15)$$

The emulator predicts $(\hat{a}, \hat{t}, \hat{s})$. The *constrained retrieval* replaces $\hat{t}$ by $\max\{\hat{t}, 0\}$ and $\hat{s}$ by its projection onto $[0, S]$, neither of which increases $|e_t|$ or $|e_s|$, and sets

$$\bar{\rho} = \Pi_{[0, R]} \left(\frac{L - \hat{a}}{\hat{t} + \hat{s}(L - \hat{a})} \right) \quad (16)$$

when the denominator is positive, and otherwise returns a prescribed value in $[0, R]$ and records a failure. It is an estimator in its own right and is reported beside the unconstrained inverse and its failure counts, never in place of them. The *retrieval-weighted component error*, computed after the projections, is

$$e_R = |e_a| + R|e_t| + R^2 t |e_s|, \quad \varepsilon_R^2 = \mathbb{E}_P[e_R^2], \quad (17)$$

which is at most $e = |e_a| + |e_t| + t|e_s|$ since $R \leq 1$; ε_R is not the mean relative radiance error of the tables. Put $F_t(\tau) = \mathbb{P}(t \leq \tau)$.

Theorem 6 (error transfer with constant one). *Under (15), for the constrained retrieval (16):*

1. $|\bar{\rho} - \rho| \leq \min\{R, e_R/t\} \leq \min\{R, e/t\}$ at every entry, and no universal constant below one is possible.

2. For every $\tau > 0$,

$$\mathbb{E}_P |\bar{\rho} - \rho|^2 \leq \mathbb{E}_P \min\left\{R^2, \frac{e_R^2}{t^2}\right\} \leq R^2 F_t(\tau) + \mathbb{E}_P \left[\frac{e_R^2}{t^2} \mathbf{1}\{t > \tau\}\right] \leq R^2 F_t(\tau) + \frac{\varepsilon_R^2}{\tau^2}.$$

3. If $F_t(u) \leq Au^\beta$ for $0 < u \leq u_0$ with some $\beta > 0$, then for $\varepsilon_R \leq u_0^{(\beta+2)/2}$, $\mathbb{E}_P |\bar{\rho} - \rho|^2 \leq (R^2 A + 1) \varepsilon_R^{2\beta/(\beta+2)}$; a component-error bound $\varepsilon_{R,n} = O(n^{-r})$ yields $\|\bar{\rho}_n - \rho\|_{L^2(P)} = O(n^{-r\beta/(\beta+2)})$.

4. If $t \geq t_0 > 0$ P -almost surely, then $\|\bar{\rho} - \rho\|_{L^2(P)} \leq \varepsilon_R/t_0$.

Proof in Appendix A. The middle expression in (ii) keeps the association between the errors and the transmission and is the one the tables evaluate; the last needs no power law. Both are computable on a finite test table; (iii) is a population statement whose tail hypothesis a finite table cannot establish down to $t = 0$. The exponent in (iii) cannot be improved from its hypotheses.

Proposition 7 (the exponent is attained). *For every $\beta > 0$, every $0 < c < r_0 < R$ and every $\tau \in (0, 1)$ there are an evaluation distribution and a predictor satisfying (15) with $S = 0$ and $F_t(u) = u^\beta$ on $(0, 1)$, namely $\rho = r_0$, $a = s = 0$, $\hat{t} = t$, $\hat{s} = 0$ and $\hat{a} = ct \mathbf{1}\{t \leq \tau\}$, for which $\bar{\rho} = r_0 - c \mathbf{1}\{t \leq \tau\}$,*

$$\begin{aligned} \varepsilon_R^2 &= \frac{c^2 \beta}{\beta + 2} \tau^{\beta+2}, & \mathbb{E}_P |\bar{\rho} - \rho|^2 &= c^2 \tau^\beta, \\ \mathbb{E}_P |\bar{\rho} - \rho|^2 &= c^{4/(\beta+2)} \left(\frac{\beta + 2}{\beta}\right)^{\beta/(\beta+2)} \varepsilon_R^{2\beta/(\beta+2)}. \end{aligned}$$

Proof in Appendix A. This is sharpness of the transfer inequality for the physical inverse, not a minimax statement over retrieval algorithms. What decides the rate is where the errors sit relative to the transmission.

Theorem 8 (conditional error profile). *Under (15), suppose $\mathbb{E}_P[e_R^2 | t] \leq \sigma^2 t^{2\nu}$ P -almost surely for some $0 \leq \nu \leq 1$. Then*

$$\mathbb{E}_P |\bar{\rho} - \rho|^2 \leq \mathbb{E}_P \min\{R^2, \sigma^2 t^{-2(1-\nu)}\}.$$

Under the tail hypothesis of Theorem 6(iii) and $\nu < 1$, as $\sigma \rightarrow 0$,

$$\mathbb{E}_P |\bar{\rho} - \rho|^2 = \begin{cases} O(\sigma^{\beta/(1-\nu)}), & \beta < 2(1-\nu), \\ O(\sigma^2 \log(1/\sigma)), & \beta = 2(1-\nu), \\ O(\sigma^2), & \beta > 2(1-\nu), \end{cases}$$

and at $\nu = 1$, $\mathbb{E}_P |\bar{\rho} - \rho|^2 \leq \sigma^2$ with no tail hypothesis. In the model of Proposition 7 the predictor $\hat{a} = a + \min\{ct, \sigma t^\nu\}$ has $e_R \leq \sigma t^\nu$ and $|\bar{\rho} - \rho| = \min\{c, \sigma t^{\nu-1}\}$, and attains each order.

Proof in Appendix A. At $\nu = 0$ the theorem recovers the uniform-error regimes under the weaker hypothesis of a bounded conditional second moment; at $\nu = 1$ an emulator whose conditional root-mean-square error decreases proportionally to the transmission loses no rate. The central diagnostic is therefore the profile $t \mapsto \mathbb{E}[e_R^2 | t]$ and not only the tail $t \mapsto F_t(t)$: the same overall error concentrated in the opaque entries gives the worst-case loss, and proportional to the transmission gives none.

8.4 Perturbing a learned-mean kernel predictor

Fix the training sample and write a corrected predictor as $p(x) = h(x) + k_x^\top H^{-1}(Y - M)$ with $H = K + \lambda I$ positive definite, $Y \in \mathbb{R}^{n \times q}$ the targets and $M \in \mathbb{R}^{n \times q}$ the training-row means used as correction labels. Let a second fitted system be $\tilde{p}(x) = \tilde{h}(x) + \tilde{k}_x^\top \tilde{H}^{-1}(\tilde{Y} - \tilde{M})$ with $\tilde{H}$ symmetric positive definite; both systems may depend on Y through the mean, the features, the selected ridge and any kernel approximation. Put $B = H^{-1/2}(\tilde{H} - H)H^{-1/2}$, whose least eigenvalue exceeds -1 , $Q = (I + B)^{-1}$, $\gamma_Q = \|Q\|_{\text{op}} = 1/(1 + \lambda_{\min}(B))$, $\eta_Q = \|Q - I\|_{\text{op}} = \max_i |b_i|/(1 + b_i)$ over the eigenvalues b_i of B , and

$$g = H^{-1/2}k_x, \quad \tilde{g} = H^{-1/2}\tilde{k}_x, \quad v = \tilde{g} - g, \quad W = H^{-1/2}(Y - M), \quad U = H^{-1/2}(\tilde{M} - M).$$

A conditioning argument bounds such a difference by the condition number of H times a norm of the change [109, 107, 110]. The statement below instead keeps the residual action W and the label shift U separate, which is what makes it readable on data.

Proposition 9 (same-data perturbation). *1. The exact difference is*

$$\tilde{p}(x) - p(x) = \tilde{h}(x) - h(x) + v^\top W - (Q\tilde{g})^\top (BW + U).$$

2. $\|\tilde{p}(x) - p(x)\|_2 \leq \|\tilde{h}(x) - h(x)\|_2 + \|W^\top v\|_2 + \|Q\tilde{g}\|_2 \|BW + U\|_{\text{op}}.$
3. $\|\tilde{p}(x) - p(x)\|_2 \leq \|\tilde{h}(x) - h(x)\|_2 + \|W\|_F (\gamma_Q \|v\|_2 + \eta_Q \|g\|_2) + \gamma_Q \|U\|_F \|\tilde{g}\|_2.$
4. *If $\tilde{H} = H + sI$ with $s \geq 0$ and the kernel and the means unchanged, then $\|\tilde{p}(x) - p(x)\|_2 \leq \|g\|_2 \|W\|_F s / (\lambda_{\min}(H) + s).$*

Proof in Appendix A. Only positive definiteness of $\tilde{H}$ is needed. A large positive eigenvalue of B , as can arise from increasing the ridge, does not invalidate the bound. The bound can nonetheless be numerically large, especially when the least eigenvalue approaches -1 . Item (ii) keeps the interaction between the perturbation and the residual: with $v = 0$, $U = 0$ and $BW = 0$ the kernel correction does not move, however large $\|B\|$ is; the full predictor is unchanged as well when $\tilde{h}(x) = h(x)$. The bounds are stated in the residual W , not in Y , they use no independence between the representation, the mean, the ridge and Y , and they transfer between fitted predictors without being generalisation statements.

For component j the physical target at band b is $f_{j,b} = \mu_{j,b} + \sum_\ell (A_j)_{b\ell} z_{j,\ell} + \pi_{j,b}$, with A_j the standardiser scale times the retained PCA basis and π_j the truncation residual, and the pipeline predicts the retained coordinates. For a frozen realised system $p_\ell(x) = h_\ell(x) + a_x^\top (y_\ell - h_{\ell,X})$ with $y_\ell = f_{\ell,X} + \varepsilon_\ell$, the conditional RKHS evaluation bound [8, 12] gives

$$|p_\ell(x) - f_\ell(x)| \leq B_\ell \tilde{P}_\lambda(x) + |a_x^\top \varepsilon_\ell|, \tag{18}$$

where, writing $k_x = k(X_n, x)$ and $a_x = (K + \lambda I)^{-1}k_x$,

$$\tilde{P}_\lambda(x)^2 = k(x, x) - 2a_x^\top k_x + a_x^\top K a_x = k(x, x) - k_x^\top (K + \lambda I)^{-1} k_x - \lambda \|a_x\|_2^2.$$

This follows by applying the RKHS Cauchy–Schwarz inequality to the residual and $k(x, \cdot) - \sum_i (a_x)_i k(x_i, \cdot)$. Here B_ℓ is a bound on the residual RKHS norm, a hypothesis and not an observed

certificate, and the second term zero for deterministic simulator targets, which is the form used for EMIT. A perturbed system adds the bound $\Delta_\ell(x)$ of Proposition 9. If $|\widehat{z}_{j,\ell} - z_{j,\ell}| \leq u_{j,\ell}$, then

$$|\widehat{f}_{j,b} - f_{j,b}| \leq |\pi_{j,b}| + \sum_{\ell} |(A_j)_{b\ell}| u_{j,\ell}, \quad (19)$$

and a change of basis or standardiser between two systems adds its own reconstruction term. Assembling $e_R = |e_a| + R|e_t| + R^2t|e_s|$ from (19) for the three physical components and applying Theorem 6 closes the chain. If the inverse is evaluated at $L_{\text{obs}} = L + \eta$, the proof of Theorem 6 goes through with e_a replaced by $e_a - \eta$, so $|\bar{\rho} - \rho| \leq \min\{R, (|e_a - \eta| + R|e_t| + R^2t|e_s|)/t\}$. The theorem is an exact-arithmetic statement. In floating point, the added term $\rho t/q$ can be lost when it is small relative to the spacing of representable numbers near a . Its loss depends on ρ , q , magnitude and rounding, not on a universal threshold $t < 2^{-52}a$. A rounded radiance may then give zero for $L - a$ even with the tabulated components. The archived operational diagnostics do not implement an arbitrary-precision verification of these entries, so a numerical violation count is not a proof of the exact-arithmetic inequality.

8.5 The stacking objective

Affine fitting predictions and projected retrieval. Let w denote simplex weights, with one simplex for each component being combined. Write $\tilde{a}_w$, $\tilde{t}_w$ and $\tilde{s}_w$ for the affine combinations used in fitting, and $\tilde{e}_a = \tilde{a}_w - a$, $\tilde{e}_t = \tilde{t}_w - t$, $\tilde{e}_s = \tilde{s}_w - s$ for their errors. Retrieval uses $\hat{t}_w = \max\{\tilde{t}_w, 0\}$ and $\hat{s}_w = \Pi_{[0,S]}\tilde{s}_w$, followed by (16). Base predictions may instead be projected before combination, provided the values entering the fitting objective remain affine in w . In either case, projection during retrieval does not increase flux or albedo error under (15).

For a fixed finite threshold $\tau > 0$, the total-flux raw-error surrogate is

$$J_\tau(w) = \mathbb{E}_P \left[\frac{\tilde{e}_a^2 + R^2\tilde{e}_t^2 + R^4t^2\tilde{e}_s^2}{(t \vee \tau)^2} \right]. \quad (20)$$

It is a convex quadratic: the errors are affine in the weights and the entry weights are nonnegative and independent of w . Its empirical version replaces the expectation by the validation average. Theorem 6 and Cauchy–Schwarz imply

$$\mathbb{E}_P |\bar{\rho}_w - \rho|^2 \leq R^2 F_t(\tau) + 3J_\tau(w). \quad (21)$$

Indeed, projection gives $e_R \leq |\tilde{e}_a| + R|\tilde{e}_t| + R^2t|\tilde{e}_s|$; split the expectation at $t = \tau$ and apply the three-term square bound on $\{t > \tau\}$. This inequality holds for every feasible w , not only an optimizer.

The implemented separable surrogate. The diagnostic driver `code/joint_stack.py` combines Y_2 and Y_3 with separate weight vectors. It projects each base albedo onto $[0, S]$ before fitting but leaves the base fluxes unprojected. With $\tilde{e}_j = \tilde{Y}_j - Y_j$ for $j = 2, 3$, its finite-threshold fitting objective is

$$J'_\tau(w) = \mathbb{E}_P \left[\frac{\tilde{e}_a^2 + R^2\tilde{e}_2^2 + R^2\tilde{e}_3^2 + R^4t^2\tilde{e}_s^2}{(t \vee \tau)^2} \right]. \quad (22)$$

This is also a convex quadratic. It omits the cross term $2R^2\tilde{e}_2\tilde{e}_3$ present in J_τ . Crucially, the flux floor is applied *after fitting*, to the combined total flux when retrieval is scored. It is not inside the fitted loss. Since $t \geq 0$,

$$|\max\{\tilde{Y}_2 + \tilde{Y}_3, 0\} - t| \leq |\tilde{e}_2 + \tilde{e}_3| \leq |\tilde{e}_2| + |\tilde{e}_3|.$$

The same argument now uses four squared terms and gives

$$\mathbb{E}_P |\bar{\rho}_w - \rho|^2 \leq R^2 F_t(\tau) + 4J'_\tau(w). \quad (23)$$

The factor-three statement is therefore not the bound for the reported separable fit. Both statements require the physical hypotheses on the distribution being averaged; full-table statistics that include entries outside those hypotheses are not certificates of either bound.

Putting projection *inside* a squared fitting loss is a different problem and need not preserve convexity. On $0 \leq w \leq 1$, for example, $g(w) = (\max\{2w - 1, 0\} - 1)^2$ satisfies $g(1/2) = 1 > (g(0) + g(1))/2 = 1/2$. This counterexample does not apply to the raw-error quadratic actually fitted by the driver.

Numerical weights and threshold convention. The historical driver solves a nonnegative least-squares problem with a heavily weighted row penalizing $\sum_h w_h - 1$, then normalizes the nonnegative weights to sum to one. This yields feasible simplex weights but is not an exact solution of the equality- constrained quadratic program. Its mean-relative-norm arm uses six iteratively reweighted least-squares sweeps from the squared-loss solution; no convergence or global-optimality certificate is recorded. We therefore describe these as approximate fits, without changing the archived predictions or attributing a new optimizer to them.

The driver expresses a finite threshold as $\tau = uT_{\text{train}}$, where $T_{\text{train}} > 0$ is a training-data flux scale. Table 11 reports u , not a flux in physical units. Its legacy `inf` option is an unweighted physical-component squared-error baseline; it is *not* the limit of (22) as $\tau \rightarrow \infty$. Its stored zero coverage field is a sentinel, not a measured tail probability. The threshold ladder is a sensitivity analysis. Selecting a deployment threshold requires an advance coverage rule; minimizing a diagnostic bound over test thresholds does not establish validation-selected retrieval performance.

The stack candidates in this study are fitted on validation rows already used for other model decisions. Proposition 5 does not certify these fitted weights. Moreover, the broader study used exploratory test feedback. Fixing a particular fit before its test evaluation does not make the study an untouched confirmation experiment.

9 Retrieval error and the transmission profile

The transfer theorem identifies the ratio of component error to transmission as a retrieval diagnostic. The conditional-profile theorem additionally requires a bound on $\mathbb{E}[e_R^2 \mid t]$ for a population and, for rate statements, an appropriate sequence of errors. These are distinct uses of the theory. This section describes the archived finite-table diagnostics without interpreting a fitted log-log slope as an asymptotic learning rate.

Table 9 uses the four common float64 prediction records at seeds 104–107. Each contains the same nine heads on 2,331 test states and 285 bands, or 664,335 state–band entries. The separate seed-101 record contains only three heads, uses float32 predictions and explicitly opts into precision-altered scoring. It is not pooled into this comparison. The revised four-record means are a reanalysis of existing summaries, not a new training campaign. The retrieval uses $\rho = 0.7$, $R = 0.9$ and a training-derived albedo bound S .

Physical domain and numerical scope. The driver defines S as the largest finite training albedo not exceeding one. The records count 227 or 454 training entries above one, depending on the split. On seed 104, 228 of the 664,335 test *entries*, not 228 of the 2,331 states, exceed

A. Errors and finite-range profiles

head	ε_R	2ν (fit)	raw p95	constrained p95
			raw p99	constrained p99
isotropic kernel	0.0886±0.0331	0.16±0.11	0.696±0.000 0.97±0.02	0.266±0.018 0.70±0.00
network	0.1650±0.0177	0.20±0.16	0.670±0.026 0.99±0.14	0.268±0.111 0.70±0.00
network ensemble	0.1085±0.0254	0.18±0.14	0.625±0.013 0.86±0.08	0.239±0.078 0.70±0.01
network + residual	0.0698±0.0373	0.21±0.14	0.489±0.054 0.89±0.01	0.200±0.000 0.70±0.00
ensemble + residual	0.0690±0.0381	0.22±0.14	0.464±0.033 0.88±0.01	0.200±0.000 0.70±0.00
feature kernel	0.0514±0.0454	0.26±0.16	0.075±0.014 0.89±0.05	0.076±0.015 0.70±0.00
concatenated features	0.0634±0.0680	0.28±0.16	0.060±0.004 0.92±0.03	0.061±0.004 0.70±0.00
coordinate selection	0.0721±0.0648	0.25±0.15	0.098±0.031 0.85±0.03	0.099±0.032 0.70±0.00
convex stack	0.0600±0.0661	0.25±0.14	0.143±0.015 1.09±0.05	0.151±0.015 0.70±0.00

B. Failure counts and legacy diagnostics

head	failure [%] upper clip [%]	$\mathbb{E} \min\{R^2, e_R^2/t^2\}$	legacy violations	diagnostic bound / MSE
isotropic kernel	1.09±0.03 3.56±0.07	0.064±0.000	0	3.9±0.2
network	1.31±1.00 2.93±0.43	0.056±0.003	0	3.4±0.7
network ensemble	0.61±0.33 2.56±0.92	0.051±0.002	0	3.7±0.9
network + residual	0.82±0.02 2.51±0.12	0.044±0.001	0	4.0±1.2
ensemble + residual	0.85±0.01 2.43±0.07	0.043±0.001	0	3.9±1.3
feature kernel	0.71±0.02 1.53±0.10	0.031±0.001	0	5.1±2.3
concatenated features	0.72±0.02 1.49±0.02	0.030±0.000	0	5.3±2.4
coordinate selection	0.73±0.02 1.61±0.13	0.033±0.002	0	4.7±2.0
convex stack	0.84±0.05 1.78±0.08	0.036±0.001	0	4.4±2.0

Table 9: Transmission diagnostics on the common four-record float64 cohort, seeds 104–107. Every cell with a spread is a mean $\pm$ sample standard deviation across these four records. All reflectance errors in this table are in *reflectance units*, not percentage points. Panel A gives the retrieval-weighted error, fitted profile slope, and raw/constrained percentiles. Panel B retains failure and clipping rates and the archived bound-related diagnostics. The legacy violation count tests only the stored $s \leq S$ mask; the all-band diagnostic expression and its ratio to MSE are not certified bounds on a domain containing nonpositive transmission or out-of-range albedo.

S ; the other three records report no exceedance. There are also 203, 6, 3 and 3 nonpositive-transmission entries on seeds 104, 105, 106 and 107, respectively. The stored pointwise check restricts to $s \leq S$ but does not enforce the full hypothesis $t > 0$, $0 \leq s \leq S$, nor does it recompute the forward/inverse pair in arbitrary precision. Its zero reported violations are therefore implementation diagnostics, not a verification of every premise of Theorem 6. The reported expectations and MSE ratios are full-table averages. Recomputing all of them on a common fully admissible domain requires the sample-level arrays, which are not public.

Observed error profiles. On the common cohort, the feature kernel has $\varepsilon_R = 0.0514$ and the single network 0.1650. Mean fitted profile slopes 2ν range from 0.16 to 0.28. The raw 95th-percentile error is 0.075 for the single-member feature kernel and 0.060 for concatenated features, compared with 0.143 for the campaign stack. Their constrained 95th percentiles are 0.076, 0.061 and 0.151. The diagnostic middle-bound/MSE ratios range from about 3.4 to 5.3 across heads; this is an observed numerical comparison, not a guarantee on the full-table distribution.

About five percent of entries have transmission below 3×10^{-4} of the training flux scale. Among positive transmissions, the one-percent quantile is of order 10^{-10} of that scale. The recorded share of squared retrieval-weighted component error below 0.03 of the scale is less than a thousandth for each head. Together, these measurements are consistent with amplification of small absolute errors by weak transmission. They do not make the heads irrelevant to the inverse tail: the location, sign and dependence of their errors still matter, as the exact identities show.

The finite fitted slopes establish neither a population tail law down to zero nor a conditional-error envelope uniform over training sizes. In particular, they do not establish that the worst-case exponent in Theorem 6(iii) is attained by these emulators. On a prespecified admissible passband $t \geq t_0 > 0$, the deterministic bound in item (iv) applies directly. Choosing that passband changes the evaluation domain and must be accompanied by coverage and failure counts; it is not a way to replace the all-band diagnostic with an unreported easier task.

10 Representation, correction and combination

Concatenated features. Table 10 retains the four-split follow-up to the original three-member feature experiment. It reports width-2000 members and splits 104–107. Its campaign JSON inputs are absent from the public repository, although the related float64 transmission summaries are present. The table is therefore a retained historical aggregate, not a newly regenerated campaign comparison.

The concatenated-feature kernel has mean radiance error 0.0688%, compared with 0.0764% for the single-member feature kernel. The paired gain is 0.0076 ± 0.0053 percentage points, positive on three splits and -0.0002 on the fourth. The mean within-split 95th-percentile reflectance error is 5.99 points, compared with 7.51 for single-member features and 14.35 for the campaign stack. Thus this aggregate shows a tail improvement from concatenation, not a universal claim that a larger representation improves retrieval. The splits overlap and do not supply independent external validation.

Fitting objective versus retrieval metric. Table 11 retains the historical joint-stacking aggregate. Its original header identifies five records, but the per-record stack JSONs are absent, so its cohort membership, solver diagnostics and per-head precision cannot be independently

seed	head	radiance [%]	median [pp]	p95 [pp]	gain [pp]
104	feature kernel	0.0924	0.044	9.25	–
104	concatenated features	0.0810	0.025	6.25	+0.0114
104	campaign stack	0.0781	0.024	12.77	–
104	network + residual	0.1843	0.115	54.71	–
105	feature kernel	0.0611	0.031	7.31	–
105	concatenated features	0.0508	0.028	6.02	+0.0103
105	campaign stack	0.0475	0.024	14.50	–
105	network + residual	0.1349	0.087	42.61	–
106	feature kernel	0.0789	0.043	7.67	–
106	concatenated features	0.0702	0.035	6.28	+0.0087
106	campaign stack	0.0687	0.034	16.29	–
106	network + residual	0.1608	0.108	51.84	–
107	feature kernel	0.0731	0.042	5.82	–
107	concatenated features	0.0732	0.044	5.41	-0.0002
107	campaign stack	0.0833	0.043	13.84	–
107	network + residual	0.1476	0.103	46.58	–
mean	feature kernel	0.0764±0.0130	0.040±0.006	7.51±1.41	–
mean	concatenated features	0.0688±0.0128	0.033±0.009	5.99±0.40	0.0076±0.0053
mean	campaign stack	0.0694±0.0158	0.031±0.009	14.35±1.48	–
mean	network + residual	0.1569±0.0211	0.103±0.012	48.93±5.40	–

Table 10: Historical four-split concatenated-feature comparison. Radiance is a mean relative spectral error in percent; reflectance median and 95th-percentile errors are in percentage points. The last column is the paired radiance gain of concatenated features over single-member features. The mean row for each head averages the four within-split quantities. The generating campaign JSONs are not present in the public archive, so this table preserves the recorded values rather than replacing them with another cohort.

reconstructed. It must not be treated as a matched extension of the four-record float64 table above. Inspection of the archived driver does establish the objective and projection sequence described in Section 8.5.

In this retained aggregate, the mean-relative-norm stack has radiance error 0.0754% and constrained 95th-percentile error 0.141, while the row-relative squared-error stack gives 0.1130% and 0.180. The finite-threshold J'_τ fits give radiance errors of approximately 0.22–0.23% for $u \leq 0.1$, 0.1491% at $u = 0.3$, and 0.1175% at $u = 1$. The unweighted baseline gives 0.1262%. These figures illustrate that the fitting loss and reporting metric need not favor the same combination. They do not establish a globally optimal threshold or an out-of-sample benefit of minimizing the surrogate. The factor-four transfer bound remains a statement about any feasible weights on a distribution satisfying its physical hypotheses.

Kernel choice on the state and on learned features. Table 12 is regenerated from twelve public component/seed records, covering four components and seeds 101–103. The spectral mixture and Matérn-5/2 have similar mean errors on Y_1 (0.089%). The spectral mixture has lower mean errors on Y_2 and Y_3 (0.237% and 0.099%) than the other tested raw-state kernels. The tested first-order additive kernel has much larger errors; this does not prove that the underlying simulator components cannot be approximated by an additive model under a different specification. Albedo errors for the nonadditive kernels remain between roughly 2.3% and 3.5%.

The Matérn-5/2 feature kernel gives errors of 0.105%, 0.211%, 0.114% and 2.303% on Y_1 – Y_4 . It improves on the best tested raw-state mean on some components, but not all. The convex stack has the lowest component means in this particular comparison (0.066%, 0.156%, 0.077%, 1.956%). This finding is compatible with the adverse reflectance-tail comparisons: the component metric does not impose the same weighting as the nonlinear inverse.

Network regularization under the tested budget. Table 13 uses three public seed records for a control, a weight-decay grid and an early-stopping grid. The validation-selected weight decay gives a similar standalone component mean to the control (2.27%); the selected early-stopping configuration is less accurate and more variable ($2.72 \pm 0.45\%$). Residual correction reduces the component means to 1.88–1.95%, and feature kernels to 0.95–0.99%, with radiance errors near 0.15%. The input reference here is *isotropic*, not the coordinate-searched input-scaled kernel of the primary table. These results do not establish that regularization is generally ineffective, only that the tested changes do not produce comparable gains under this budget.

Perturbation and ridge-path diagnostics. The extended tables in Appendix B keep all perturbation rows at a readable size. Feature-perturbation rows are regenerated from eight public records; the fourteen state-input rows are retained from their historical table because their JSON is absent. The residual-action bound is often substantially smaller than the spectrum-only bound. Nevertheless, its ratio to actual movement can be tens, thousands or much larger. Positive definiteness can make a bound applicable without making it a useful application tolerance. The reported feature-record identity and bound checks are numerical checks on fitted systems, not generalization tests.

Table 16 reports a 4,000-row fit and four components at seed 101. At the smallest ridge on input-space Y_1 , the effective degrees of freedom are 3,841 and the noise factor is 0.924. The same-half and split-half PCA errors are 0.079% and 2.465%, whereas the physical-unit test errors for the corrected and standalone kernels are 0.2168% and 0.1754%. These columns use different

head / fit	radiance [%]		constrained p95	
			ε_R constrained p99	raw p95
network	0.8609±0.1051	0.1664±0.0156	0.254±0.101	0.677±0.027
			0.70±0.00	
			0.200±0.000	
network + residual	0.1560±0.0184	0.0654±0.0338	0.70±0.00	0.486±0.047
			0.077±0.013	
			0.70±0.00	
feature kernel	0.0755±0.0114	0.0472±0.0404	0.70±0.00	0.076±0.012
			0.266±0.018	
			0.70±0.00	
isotropic kernel	0.2454±0.0106	0.0886±0.0331	0.70±0.00	0.696±0.000
			0.239±0.078	
			0.70±0.01	
network ensemble	0.5050±0.0592	0.1085±0.0254	0.200±0.000	0.625±0.013
			0.70±0.00	
			0.061±0.004	
ensemble + residual	0.1379±0.0157	0.0690±0.0381	0.70±0.00	0.464±0.033
			0.061±0.004	
			0.70±0.00	
concatenated features	0.0688±0.0128	0.0634±0.0680	0.141±0.033	0.060±0.004
			0.70±0.00	
			0.70±0.00	
stack, mean relative norm	0.0754±0.0116	0.0573±0.0575	0.180±0.044	0.133±0.029
			0.70±0.00	
			0.135±0.060	
stack, row-relative square	0.1130±0.0454	0.0518±0.0414	0.70±0.00	0.281±0.122
			0.103±0.055	
			0.69±0.01	
stack, J'_τ , $u = 0.001$	0.2203±0.1042	0.0633±0.0298	0.107±0.053	0.213±0.227
			0.69±0.01	
			0.107±0.053	
stack, J'_τ , $u = 0.003$	0.2159±0.1100	0.0736±0.0522	0.59±0.22	0.182±0.238
			0.107±0.053	
			0.59±0.22	
stack, J'_τ , $u = 0.01$	0.2303±0.1023	0.0761±0.0508	0.110±0.052	0.185±0.236
			0.59±0.22	
			0.107±0.053	
stack, J'_τ , $u = 0.03$	0.2304±0.1024	0.0761±0.0508	0.117±0.048	0.185±0.235
			0.59±0.22	
			0.110±0.052	
stack, J'_τ , $u = 0.1$	0.2298±0.1013	0.0739±0.0463	0.69±0.03	0.188±0.235
			0.156±0.032	
			0.117±0.048	
stack, J'_τ , $u = 0.3$	0.1491±0.0508	0.0615±0.0468	0.70±0.00	0.183±0.196
			0.168±0.039	
			0.156±0.032	
stack, J'_τ , $u = 1.0$	0.1175±0.0422	0.0553±0.0469	0.70±0.00	0.152±0.032
			0.168±0.039	
			0.70±0.00	
stack, unweighted square	0.1262±0.0406	0.0559±0.0420	0.70±0.00	0.194±0.095

Table 11: Historical joint-stacking aggregate at $\rho = 0.7$, retained without refitting. The finite-threshold rows use the raw, separable quadratic J'_τ of (22), not the total-flux J_τ . Here $u = \tau/T_{\text{train}}$. The unweighted row is a separate component-squared-error baseline, not a threshold limit. Radiance is in percent; reflectance percentiles are in reflectance units. Weights were fitted approximately using the routines described in Section 8.5. The five-record header does not establish per-head counts or a common precision cohort without the missing generating JSONs.

head	Y_1 [%]	Y_2 [%]	Y_3 [%]	Y_4 [%]
Matérn-1/2	0.179±0.008	0.672±0.003	0.418±0.012	2.326±0.103
Matérn-3/2	0.092±0.002	0.391±0.008	0.207±0.016	2.481±0.043
Matérn-5/2	0.089±0.017	0.434±0.007	0.203±0.003	2.746±0.082
Gaussian	0.092±0.002	0.525±0.113	0.228±0.051	3.502±0.501
spectral mixture (4)	0.089±0.010	0.237±0.014	0.099±0.013	3.468±0.263
NNGP, ReLU	0.175±0.001	0.674±0.015	0.350±0.009	2.548±0.057
NTK, ReLU	0.263±0.002	1.055±0.015	0.621±0.014	2.321±0.048
additive Matérn-5/2	4.091±0.093	20.266±1.035	11.430±0.237	4.413±0.077
ARD by kernel flow	0.322±0.015	0.257±0.019	0.199±0.005	2.418±0.114
ARD by marginal likelihood	0.281±0.039	0.341±0.042	0.183±0.011	2.702±0.027
convex kernel sum	0.087±0.007	0.253±0.020	0.099±0.013	2.557±0.099
network	0.819±0.027	2.289±0.019	0.910±0.057	3.482±0.085
network + residual kernel	0.523±0.007	1.337±0.026	0.588±0.035	3.055±0.121
features, Matérn-5/2	0.105±0.006	0.211±0.016	0.114±0.008	2.303±0.216
features, Gaussian	0.126±0.007	0.262±0.018	0.138±0.014	2.426±0.214
features, NTK	0.152±0.005	0.307±0.015	0.156±0.007	2.150±0.059
features, ARD	0.116±0.007	0.214±0.014	0.111±0.004	2.217±0.207
coordinate selection	0.090±0.010	0.360±0.008	0.105±0.010	2.797±0.447
convex stack of the heads	0.066±0.003	0.156±0.010	0.077±0.004	1.956±0.037
ridge stack of the heads	0.087±0.005	0.288±0.031	0.087±0.005	2.310±0.071

Table 12: Ten input-space kernels, feature kernels and combination rules on the EMIT state inputs, with rank-64 outputs. Entries are component mean relative spectral errors in percent, averaged over seeds 101–103 with sample standard deviations. The network has width 512 and depth 3 with a 150-epoch budget. Kernels include the Matérn family, Gaussian, spectral mixture [72], NNGP [93, 91], NTK [94], additive kernel [73], and learned input scales [58, 8]. Validation fitting and the reported test metric are not themselves population optimality guarantees.

head	alone	residual correction	feature kernel
	component / radiance	component / radiance	component / radiance
isotropic kernel	0.941±0.109		
	0.284±0.078		
	2.266±0.031	1.903±0.055	0.955±0.038
network, control	0.809±0.029	0.617±0.033	0.153±0.052
	2.269±0.027	1.880±0.072	0.948±0.048
network, weight decay	0.820±0.014	0.618±0.040	0.154±0.048
	2.716±0.445	1.949±0.073	0.986±0.032
network, early stopping	1.293±0.373	0.789±0.127	0.147±0.054

Table 13: Regularization comparison, seeds 101–103. Each cell gives mean component relative error [percent] / radiance relative error [percent], with sample standard deviations. The width-512, depth-3 network has a 150-epoch cap; validation selects among three runs per arm. The weight-decay grid is 10^{-5} , 10^{-4} , 10^{-3} and the early-stopping patience grid is 3, 5 and 10 epochs [87, 88]. The reference row is the isotropic kernel from this campaign.

metrics and evaluation samples and should not be subtracted as like-for-like gains. At $\gamma = 10^{-2}$ the corresponding physical-unit test errors are 0.5095% and 5.7578%: the fitted mean retains predictive information when the kernel correction is strongly regularized. The ridge path is descriptive; Proposition 4 does not guarantee this test-error ordering.

11 Synthetic reference for the transfer theorem

Table 17 is a manufactured-data check, not additional evidence about EMIT. The revision reruns the existing construction with two million samples per β , seed zero, at $\beta = 0.5, 1, 2, 3, 4$, and archives the resulting JSON. The average-error slopes are 0.200, 0.333, 0.498, 0.597 and 0.664, compared with theoretical limits 0.200, 0.333, 0.500, 0.600 and 0.667. The finite-ladder uniform-family slopes are compared with the analytic curve on the *same* ladder, rather than only its limiting exponent. At $\beta = 2$, the measured slope is 1.721 and the analytic finite-ladder slope 1.724; the limit is two.

The largest relative Monte Carlo discrepancy is about 1.01% for the uniform family at $\beta = 2$ and below 0.053% for the profile family against its numerical-quadrature reference. The rare-event counts determine which average-family points enter each slope fit. No pointwise violations are found across the tested predictions on the ten million sampled transmissions. This numerical agreement supplements, but does not replace, the exact proofs.

12 The matched benchmark and the transfer contrast

Correction-coefficient benchmark. Table 18 compares archived scores under the standard and out-of-distribution protocols of [4], with and without low-fidelity 6S coefficients as inputs. The published reference is not retrained here, and the comparison does not control for all differences in optimization or computational budget. The claim is therefore a comparison with the published values under the stated protocol, not an experimentally isolated architectural advantage.

The state-only residual-corrected network gives standard-split RMSE 0.00581 ± 0.00004 , MAE 0.00151, uniform-average $R^2 = 0.9952$ and SMAPE 3.54%, compared with published values 0.00619, 0.00186, 0.9947 and 3.91%. With 6S inputs its means are 0.00578, 0.00148, 0.9954 and

3.19%. On the specified out-of-distribution split, the state-only means are 0.00620, 0.00252, 0.9967 and 3.37%, compared with published values 0.01022, 0.00544, 0.9942 and 5.40%. With 6S inputs the means become 0.00596, 0.00232, 0.9969 and 2.92%. The spreads describe three seeds on the standard protocol and three initializations on the out-of-distribution protocol.

The R^2 convention is material. The reference comparison weights the three coefficients equally. The released rescore records instead store a variance-weighted R^2 alongside coefficientwise RMSE. The table generator reconstructs the common per-coefficient target scales from the recorded mean normalized RMSE across heads, checks them against the stored variance-weighted R^2 , and computes the uniform-average value. This reconstruction uses archived summary identities rather than rescoring public raw predictions. It now fails explicitly when that convention cannot be reconstructed, rather than silently substituting the other R^2 . The ensemble without correction is close to the corrected network, while the raw-state kernel is much less accurate in these recorded configurations. These results should be read alongside the input, approximation and tuning choices, not as a general ordering of kernels and networks.

OCO-2 coefficient and radiance objectives. Table 19 is regenerated from nine public selection records: three bands and seeds 0–2. The feature kernel trained under the flat coefficient loss gives coefficient errors of approximately 4.1%, 16.3% and 8.1% on O₂, weak CO₂ and strong CO₂, but radiance errors of 0.076%, 0.114% and 0.108%. A radiance-weighted feature construction instead gives approximately 0.031%, 0.048% and 0.064% in radiance, with larger coefficient errors. Thus the reconstruction map and evaluation metric can change which fitted representation is preferred.

The radiance stack minimizes a *row-relative squared* radiance loss, a convex quadratic because the reconstruction is affine. The reported radiance error is the mean of row-relative norms, not that squared fitting objective. The recorded solver has KKT gaps from 4.17×10^{-6} to 1.52×10^{-5} for its quadratic program; this does not certify optimality in the reported norm or on test data. Its mean radiance errors are 0.0264%, 0.0391% and 0.0511%, compared with 0.0291%, 0.0404% and 0.0536% for the reduced-coordinate squared-loss stack. Coefficient errors increase to 18.59% and 37.43% on O₂ and weak CO₂, and to 8.87% on strong CO₂. All rules use 2,000 validation and 2,000 test rows in these records. The ten-split companion comparisons in the following section are a separate campaign, not ten repetitions of this three-seed selection experiment.

13 Comparison with OCO-2

The companion paper [12] applies the same construction to the OCO-2 radiative-transfer emulation problem of Lamminpää et al. [3]: per spectral band (O₂, weak CO₂, strong CO₂), a reduced atmospheric state of twenty or twenty-four coordinates maps to forty reduced radiance coefficients, with errors on the standardized coefficients and on the reconstructed radiance, under the ten-split protocol of Section 3. The corresponding results in [12] provide a comparison at a higher input dimension.

On this higher-dimensional problem, a kernel on the state trails the network by a factor of nine, four and five on the reduced coefficients: the isotropic Matérn reaches 40.6%, 58.6% and 43.4% on the three bands against 4.4%, 16.4% and 8.2% for the network. The residual correction adds no material gain in the reported OCO-2 comparisons. Little alignment with useful kernel directions is a possible explanation, but the public records do not measure the residual’s Gram-eigenvector coefficients. The design identity in Proposition 2 alone cannot establish that

explanation on test data. The coordinate search that gives a kernel on the state its own length scales, a strong raw-input construction on the EMIT table, is not the strongest there either. Four learned metrics were run at the same ten splits on all three bands: a kernel-flow metric [58], a Mahalanobis metric under the same flow, the gradient outer product of the recursive feature machine [97], and per-input scales fitted by empirical Bayes. The last is the best of the four at every band, taking the isotropic kernel’s 40.6%, 58.6% and 44.3% on the coefficients to 18.7%, 25.3% and 16.9%, and on O₂ its radiance from 0.233% to 0.150%; the two flow metrics agree within half a point of each other at 26.9%, 29.8% and 23.9%, the recursive feature machine reads 28.2%, 47.9% and 31.3%, and an additive kernel 38.2%, 46.9% and 47.7%. The isotropic baseline agrees between the metric and baseline experiments on O₂ and weak CO₂ to two decimals; on strong CO₂ it reads 44.3% in the metric experiment and 43.4% in the baseline experiment. This band also has a spread of ± 3.4 because two of its ten splits land three points below the other eight. The learned metric reduces the isotropic error to between two fifths and a half of itself depending on the band. The feature kernel remains more accurate, with a band-dependent margin: the exact Matérn regression on the network’s frozen features reads 4.1%, 16.3% and 8.1% at the same splits, so the best state metric is 4.6 times that on O₂ but only 1.6 and 2.1 times it on the two CO₂ bands. The empirical-Bayes fit has substantial split variability: on O₂ the empirical-Bayes metric is bimodal across splits, eight of the ten landing near 17.2% and two at 24.7%, so its spread of ± 3.2 is thirty times the isotropic kernel’s ± 0.1 and its mean is not the value a single split is likely to show. The feature-kernel construction remains useful on both tasks: the exact Matérn regression on the network’s frozen features beats a ridge readout refitted on the same features at every split, on every band and under every training loss, and a per-coefficient choice among heads trained under different losses has lower mean error than the release’s stored emulator under both reported metrics. In the archived loss campaign it wins the reduced-coefficient comparison at all ten splits on each band and the radiance comparison at ten, nine and ten splits on O₂, weak CO₂ and strong CO₂, respectively. These companion comparisons are not an independent EMIT validation; their tables are in [12].

The OCO-2 retrieval records overlap the release’s test set. The reference’s data release carries 976 retrieval records, each with an atmospheric state, the full-physics monochromatic radiance at that state, its rank-40 reconstruction, and the prediction of the release’s Gaussian-process emulator at the same forty coefficients. Every one of the 976 reduced states matches a stored test input within 6.67×10^{-16} ; the records map to 974 distinct test rows, and the nearest test-row index is the same in all three bands. The records are therefore not disjoint from the stored test set and cannot serve as an independent evaluation pool, though they do provide a different target, the full-physics radiance, at those states. The state reduction was replayed independently from the release’s two dimensional-reduction files and the `x_in` and `geo_in` fields of all 976 records; the replay code and its digests are archived with the per-run records.

14 Ten further emulation configurations

Table 14 extends the comparison to ten emulation configurations. Each family uses validation-based hyperparameter selection. The models and training budgets are specified below; these comparisons assess the selected family implementations rather than the best task-specific methods available for each corpus.

The corpora are the advection equation with discontinuous initial data from the operator suite of [103], which is also the source of the structural-mechanics problem of [12]; Burgers’

equation at three viscosities from PDEBench [104]; three one-step maps from The Well [105] — a two-dimensional turbulent radiative layer, an active-matter system, and a Helmholtz staircase; the ClimSim atmospheric physics emulation task [106]; and the paired correction-coefficient corpus of the physics-guided radiative-transfer work of [4], which is a third radiative-transfer emulation problem and the largest here at 487,790 rows. That last corpus is run in both of its configurations: from the atmospheric state alone, and with the low-fidelity 6S coefficients supplied as additional inputs, which is a different and easier problem.

The configurations are these. Advection is the 200-point input/output pair of the operator suite, used at full resolution with no reduction. Burgers is PDEBench’s `1D_Burgers_Sols_Nu` release, the map from the initial condition to the solution at $t = 1$ on a grid subsampled by four to 256 cells, the input used raw and the output reduced to 128 principal coefficients. The three Well datasets are one-step maps of the full multi-channel state with 256 principal coefficients in and out: the turbulent radiative layer on its 128×384 grid (density, pressure, velocity), active matter on 256×256 (concentration, velocity, and the two tensor fields), and the Helmholtz staircase on its two pressure channels. ClimSim is the low-resolution release, 124 inputs to 128 tendencies, with 100,000 training rows drawn from the 10.1M available and validation and test taken from the release’s own validation and scoring files. The correction-coefficient corpus is the paired 6S/libRadtran Sentinel-2 set [14, 15], seven state variables and band wavelength mapping to the three libRadtran coefficients, split by atmospheric state so that all bands of a state travel together. The inputs are wavelength, solar zenith, view zenith, relative azimuth, aerosol optical depth, water vapor, ozone and elevation. The multi-fidelity configuration adds the three 6S coefficients, giving input dimensions eight and eleven, respectively. For these two configurations the error is

$$E_{0.05} = \frac{1}{N} \sum_{i=1}^N \frac{\|Y_i - \hat{Y}_i\|_2}{\max\{\|Y_i\|_2, 0.05\}}. \quad (24)$$

The same denominator floor enters validation selection and the kernel-flow objective. It limits the weight of small-norm coefficient vectors; these two rows therefore use a floored relative error rather than an unmodified relative norm.

The families are those of Section 4 with two differences forced by scale. The reference family is a ridge regression linear in whatever inputs the models receive, not the cubic of Table 1: a cubic expansion of a 256-dimensional input is not tractable, and the linear reference is used consistently for the additional configurations. And the kernel solves are exact only when the training block has at most 20,000 rows; above that they use a Nyström approximation [60] on 6,000 uniformly drawn landmarks. That cap binds on ClimSim, on the Helmholtz staircase at 20,384 rows, and on the two correction-coefficient configurations. Those entries are what a capped fit gives; they are approximate rather than full dense solves. Their errors need not bound those of the full kernel, since approximation affects both fit and regularization. The convex stack and the per-coordinate selection are formed from the kernels, the networks, the residual correction and the feature kernel; the ridge is a reference row and is not one of their members.

What each row averages differs and is stated in the table. Advection, Burgers, ClimSim and the two correction-coefficient configurations have three to five random splits. The Well’s releases carry their own train, validation and test files, which are used unchanged, so the three turbulent-radiative-layer runs share one split and differ only in network initialisation; their spread, of order 0.01 points, measures initialisation and not sampling. The active-matter and Helmholtz-staircase rows are single runs.

The relative errors vary substantially with the problem. Where the map is smooth and the

sample small — Burgers at $\nu = 0.1$ — the exact kernel is at 2.81% against the network’s 20.34%, a factor of seven, of the same kind as the five- to six-fold gap between the per-input kernel and the network on the EMIT table’s smooth components; as the viscosity falls and the solutions roughen the gap closes to 23.40% against 36.65% and then to 31.66% against 39.98%. Where the sample is large and the map rough — ClimSim, and the correction-coefficient corpus at nearly half a million rows — the network leads instead, 25.10% against 27.20% and 3.75% against 62.93%, which is the OCO-2 regime. Where the operator is linear, on advection, the linear ridge is the lowest row and every other family lands within 0.4 points of it, consistent with the linear form of the operator.

Learned metrics give particularly large improvements on the correction-coefficient corpus. On the correction-coefficient corpus from the state alone the isotropic kernel is at 62.93 ± 0.51 and a kernel-flow metric [58] takes it to 3.97 ± 0.42 , a factor of sixteen in the mean, which brings a kernel on the state to within six percent of the network on a corpus where the plain kernel was seventeen times worse. With the low-fidelity coefficients supplied as inputs the same pair reads 15.76 and 4.60, a factor of three: the easier the problem is made for the plain kernel, the less the learned metric adds. This comparison, like the EMIT and OCO-2 results, shows that the choice of input metric can substantially change a kernel baseline.

The convex stack is at or near the best member of its candidate pool on each configuration. The convex stack is the lowest row on eight of the ten configurations and the second lowest on the other two; the residual-corrected network is in the lowest two on five. Both exceptions are the linear problems, and on both the row that beats the stack is the ridge, which the stack does not contain. On advection the ridge is lower by 0.07 points while the stack, at 11.36%, matches the best member it has — the isotropic kernel, on which it puts weight 0.87 — to two decimals. On the Helmholtz staircase the ridge reaches 0.004%, sixteen times below the next row of the table and about a hundred times below the weakest member of the pool, and the stack reaches 0.066%, again a shade below its own best member, the learned-metric kernel at 0.067%, on which it puts weight 0.94. In both cases the stack does what it is asked to do inside its pool; what it cannot do is recover a family the pool excludes. The statement this supports is correspondingly narrow: a combination is a good default when no member of its own pool dominates it, and a combination can only be read against the whole comparison when the pool contains the families the comparison includes.

The comparisons have several limitations. These rows use one training budget per corpus and one architecture family, and several of these corpora have published results from methods tuned for them specifically; the comparison here is between families under a common protocol, not against the state of the art on each problem. With three to five runs, and one on two rows, the spreads support the large orderings above and not fine distinctions between adjacent families. And every kernel column here inherits the tuning protocol Section 6.1 measures: the length scale is chosen on a 6,000-row subsample and the winner refit on every row, and this difference in sample size can affect the selected hyperparameters. On the EMIT table that costs the isotropic kernel a factor 0.882; on ClimSim, where the subsample is a sixteenth of the solve, re-choosing the scale against the solve that actually runs halves it and lowers the kernel’s error by 3.6%; on a 20,000-row corpus it changes nothing. These examples show that tuning can affect the comparison. They do not imply that every reported kernel error is pessimistic or that a reported gap is an upper bound: validation retuning can also increase test error. The columns of Table 14 describe the specified training and selection procedures, not the optimum attainable by either family.

15 Discussion

The main EMIT comparison supports the use of both input scaling and learned features. The input-scaled Matérn kernel substantially reduces radiance error relative to the isotropic kernel, cubic regression and width-512 network. Residual fitting improves the network under the implemented PCA projection. The width-2000 network has a larger standalone error under the tested optimizer and training budget, yet its features support more accurate kernel predictions at every main split. Width and training duration change together, so the experiment concerns fitted representations rather than an intrinsic benefit of width.

Forward and inverse rankings nevertheless differ. The input-scaled kernel has the smallest median reflectance error in the main comparison, while the feature kernels have smaller 95th-percentile errors. Both narrow and wide stacks increase the latter tails at every split relative to their feature kernels. For the wide model, a mean radiance gain of 0.0006 percentage points accompanies an increase from 7.57 to 22.71 points in the mean within-split 95th-percentile reflectance error. Neither a single forward score nor a reflectance median summarizes this behavior. The unconditioned inverse also produces large finite outliers, so these all-band scores should not be interpreted as retrieval accuracy on a specified physical passband.

Theorem 6 gives a precise explanation of why such a reversal is possible. On its physical domain, the constrained retrieval error is at most e_R/t , capped at R , with constant one. A distributional bound separates the small-transmission mass from errors relative to transmission on the remaining entries. Proposition 7 shows sharpness of the transfer exponent for this inverse under the stated population hypotheses, not minimax optimality over all retrieval procedures. Theorem 8 shows that a conditional root-mean-square component error proportional to transmission is sufficient to retain the component-error rate. This is not a necessary condition in every distribution: its other cases also allow rate retention for sufficiently light tails. The finite transmission profiles do not establish which asymptotic regime applies to EMIT.

The distinction between fitting and scoring is equally important. The diagnostic stack fits a convex quadratic in raw component errors, with direct and diffuse flux treated separately, and projects the combined flux only for retrieval. Its deterministic transfer bound has factor four. This differs from the factor-three bound for a total-flux quadratic that includes the flux cross term. Both bounds require the physical hypotheses and errors on the evaluation distribution; neither turns a low validation objective into a test guarantee. The implemented equality-penalized NNLS and fixed-sweep IRLS routines produce approximate weights. Their recorded predictions remain empirical results, not certificates of globally optimal retrieval or forward performance.

For residual correction, the exact alignment identity and PCA decomposition identify which error terms can improve and which remain. A learning-rate claim additionally requires a uniform regression bound and control of the fitted residual in the relevant source norm; the measured learning curves do not establish these conditions. Proposition 9 similarly controls movement between two fitted systems without asserting generalization. Its residual-action bound can be much smaller than the spectrum-only bound, but several reported ratios remain very large. A finite bound is not necessarily an adequate approximation tolerance for an application.

The supplementary studies reinforce metric- and task-dependent comparisons rather than a universally preferred family. Concatenated features improve mean radiance error on three of four follow-up splits and improve their aggregate reflectance tail. The matched correction-coefficient scores favor residual-corrected networks relative to the published reference values, with the reference not retrained here. OCO-2 coefficient and radiance losses select different trade-offs; its quadratic fitting objectives must be distinguished from the mean relative norms used for reporting.

The further-corpus results concern the specified budgets, reductions and kernel approximations, not each corpus’s best available method.

Accuracy must also be considered alongside cost. A dense kernel matrix uses order n^2 storage and its factorization order n^3 operations. One $18,884 \times 18,884$ double-precision matrix occupies about 2.85 GB before workspaces and other arrays. Feature kernels require both neural evaluation and a kernel prediction against the training design; a stack evaluates every active head. No matched throughput measurement establishes a speed advantage over operational emulators such as sRTMnet [1].

The principal empirical limitation is the adaptively explored, fixed table. Overlapping partitions are not independent external validation. The principal reproducibility limitation is that public summaries do not replace raw arrays, full-precision predictions and simulator provenance. Additional transmission summaries include nonpositive-flux and albedo-out-of-range entries; their all-band diagnostic expressions are not certified population bounds. A deployment evaluation should fix the application passband and coverage requirement before comparison, separate physical failures from numerical failures, and report both the raw and constrained inverse. Within the scope actually tested, neural features and exact kernels complement one another, while a small forward gain can coexist with a substantially worse inverse-error tail.

A. Configurations

corpus	n	d	q	runs
Advection	19,000	200	200	5
Burgers $\nu = 0.1$	7,200	256	128	3
Burgers $\nu = 0.01$	7,200	256	128	3
Burgers $\nu = 0.001$	7,200	256	128	3
Radiative layer	7,200	256	256	3
Active matter	14,000	256	256	1
Helmholtz staircase	20,384	256	256	1
ClimSim	100,000	124	128	5
RT coefficients, state	487,773–487,823	8	3	5
RT coefficients, +6S	487,773–487,823	11	3	5

B. Reference, input kernels and network

corpus	linear ridge	KRR	learned-metric	network
			KRR	
Advection	11.29±0.04	11.36±0.04	11.36±0.04	11.69±0.07
Burgers $\nu = 0.1$	13.89±0.53	2.81±0.28	2.90±0.33	20.34±0.61
Burgers $\nu = 0.01$	33.68±0.28	23.40±0.54	23.58±0.50	36.65±1.95
Burgers $\nu = 0.001$	38.38±0.12	31.66±0.16	33.23±1.73	39.98±0.65
Radiative layer	31.73	30.31	30.26±0.01	32.17±0.02
Active matter	37.24	35.81	35.18	37.31
Helmholtz staircase	0.00	0.26	0.07	0.40
ClimSim	32.24±0.06	27.20±0.05	26.31±0.04	25.10±0.18
RT coefficients, state	71.54±0.59	62.93±0.51	3.97±0.42	3.75±0.25
RT coefficients, +6S	33.83±0.26	15.76±1.98	4.60±0.31	2.79±0.14

C. Ensembles, corrections and combinations

corpus	ensemble	network +	feature kernel	stack
		residual KRR		
Advection	11.50±0.05	11.50±0.05	11.64±0.07	11.36±0.04
Burgers $\nu = 0.1$	19.28±0.09	2.78±0.29	4.92±0.30	2.78±0.29
Burgers $\nu = 0.01$	35.31±0.70	23.12±0.56	29.20±0.59	23.01±0.54
Burgers $\nu = 0.001$	38.89±0.11	31.82±0.11	35.59±0.48	31.34±0.14
Radiative layer	31.58±0.02	30.40±0.01	31.08±0.01	30.12
Active matter	36.93	35.77	36.22	35.05
Helmholtz staircase	0.24	0.23	0.45	0.07
ClimSim	24.43±0.17	24.33±0.13	24.71±0.25	24.25±0.17
RT coefficients, state	3.24±0.11	3.13±0.05	3.52±0.07	3.09±0.05
RT coefficients, +6S	2.40±0.11	2.34±0.09	2.80±0.09	2.32±0.08

Table 14: Ten further configurations: dimensions and run counts (A), reference models (B), and combinations (C). Errors are mean relative spectral norms in percent, with the 0.05 denominator floor of (24) for the two RT-coefficient rows. Spreads are sample standard deviations; the radiative-layer repetitions share one split and vary initialization, and two rows are single runs. The network budget is 200 epochs at width 512, except ClimSim (60 at 512) and RT coefficients (120 at 384). The linear ridge is excluded from the stack; its Helmholtz error is 0.004%, rounded in the table. Kernel solves above 20,000 training rows use 6,000 Nyström landmarks. Further protocol and task-specific limitations are given in the text.

A Proofs

The statements are as numbered in the text.

Proof of Proposition 1. Subtract the two forward maps and use $1/\hat{q} - 1/q = \rho e_s/(q\hat{q})$ to obtain (4). Since $u = \rho D$, $\hat{D} = D + e_t + u e_s - \hat{s} e_a$ and $\hat{\rho} - \rho = (u - e_a - \rho \hat{D})/\hat{D}$, which gives (5). The triangle inequality yields $|\hat{D} - D| \leq B$ and then (6). $\square$

Proof of Proposition 2. Subtract $h + S(f - h)$ from f to obtain (9). On the design, $I - K(K + \lambda I)^{-1} = \lambda(K + \lambda I)^{-1}$, which proves (10) and its spectral form. Apply the assumed uniform inequality to $g = f - h$ for (11). The examples below distinguish this bound from actual error comparisons. $\square$

Proof of Proposition 3. For a common operator the corrected error is $a - b$, so expansion of its squared norm gives (12). The error of p_t is $a - tb$; minimize the resulting quadratic on $[0, 1]$ to obtain (13). For distinct operators, $f - h - S_1(f - h) = (I - S_0)f - [(S_1 - S_0)f + (I - S_1)h]$, after which the same expansion applies. $\square$

Proof of Proposition 4. (i) is the chain rule applied to $\hat{y} = S_\lambda y + (I - S_\lambda)h_X(y)$, and the two examples are direct substitutions. (ii) With the mean and kernel frozen the map is affine with linear part S_λ ; the trace is the sum of the eigenvalues $\mu_j/(\mu_j + \lambda)$, and the operator norm of a symmetric positive semidefinite matrix is its largest eigenvalue. (iii) $\hat{y} - f_X = -(I - S_\lambda)r + S_\lambda \varepsilon$; the cross term has zero mean, and both matrices are diagonal in the eigenbasis of K . (iv) The correction at x is $a_\lambda(x)^\top (y - h_X)$, linear in ε with coefficient vector $a_\lambda(x)$. $\square$

Proof of Proposition 5. Let $\hat{\mathcal{R}}_j$ and $\mathcal{R}_j$ be the empirical and population risks. Hoeffding's inequality [108] and a union bound give $|\hat{\mathcal{R}}_j - \mathcal{R}_j| \leq B\sqrt{\log(2M/\delta)/(2m)}$ simultaneously for all candidates with probability at least $1 - \delta$. Apply this once to the empirical minimizer and once to a population minimizer, and use empirical optimality between the two inequalities. $\square$

Proof of Theorem 6. Write $q = 1 - \rho s$, $u = L - a = \rho t/q$, $D = t/q$, $\hat{D} = \hat{t} + \hat{s}(L - \hat{a})$ and $z = \bar{\rho}$. Suppose first $\hat{D} > 0$. The map $F(v) = \hat{D}v - (L - \hat{a})$ is increasing and affine, z is the projection of its root onto $[0, R]$ and $\rho \in [0, R]$, so $(z - \rho)F(z) \leq 0$: at an interior root $F(z) = 0$; at $z = R$ the root exceeds R , so $F(R) < 0$ and $z - \rho \geq 0$; at $z = 0$ the root is negative, so $F(0) > 0$ and $z - \rho \leq 0$. Since $L - \hat{a} = u - e_a$ and, by (5), $\hat{D} = D + e_t + u e_s - \hat{s} e_a$ with $u = \rho D$,

$$F(z) = D(z - \rho) + (1 - z\hat{s})e_a + z e_t + z u e_s,$$

hence $D(z - \rho)^2 \leq |z - \rho|((1 - z\hat{s})|e_a| + z|e_t| + zu|e_s|)$ and, with $D = t/q$,

$$|z - \rho| \leq q(1 - z\hat{s})\frac{|e_a|}{t} + qz\frac{|e_t|}{t} + z\rho|e_s|.$$

Now $q(1 - z\hat{s})$ is positive and at most 1 under the hypotheses, which is all that is used here, $qz \leq R$ and $z\rho \leq R^2$, so $|z - \rho| \leq e_R/t$; and $|z - \rho| \leq R$ because both lie in $[0, R]$. If $\hat{D} \leq 0$, then $\hat{t} \leq \hat{s}(e_a - u) \leq \hat{s}|e_a|$ and $t = \hat{t} - e_t \leq |e_t| + \hat{s}|e_a|$, so $Rt \leq R|e_t| + R\hat{s}|e_a| \leq R|e_t| + |e_a| \leq e_R$; the fallback value and ρ lie in $[0, R]$, so $|z - \rho| \leq R = \min\{R, e_R/t\}$. For optimality take $s = \hat{s} = 0$, $\hat{t} = t$, $\rho = r_0$ and $\hat{a} = a + ct$ with $0 < c < r_0 < R$: then $|\bar{\rho} - \rho| = c = e_R/t$ exactly, with a positive denominator and no projection active. This proves (i). For (ii) integrate (i), split at $t = \tau$ and bound e_R^2/t^2 by e_R^2/τ^2 on $\{t > \tau\}$. For (iii), $\varepsilon_R = 0$ gives zero error by (i). Otherwise take $\tau = \varepsilon_R^{2/(\beta+2)} \leq u_0$; then $R^2 A \tau^\beta + \varepsilon_R^2/\tau^2 = (R^2 A + 1)\varepsilon_R^{2\beta/(\beta+2)}$. (iv) is (i) with $t \geq t_0$. $\square$

Proof of Proposition 7. $L = r_0 t$ and $\hat{\rho} = (r_0 t - \hat{a})/t$; $e_R = |\hat{a}| = ct \mathbf{1}\{t \leq \tau\}$; with density $\beta u^{\beta-1}$ on $(0, 1)$, $\mathbb{E}_P[t^2 \mathbf{1}\{t \leq \tau\}] = \beta \tau^{\beta+2}/(\beta + 2)$; the retrieval error equals c exactly on $\{t \leq \tau\}$, an event of probability τ^β . Eliminate τ . $\square$

Proof of Theorem 8. If $R = 0$ the statement is immediate; assume $R > 0$. Conditional on t the squared error is at most both R^2 and e_R^2/t^2 by Theorem 6(i), so $\mathbb{E}[|\bar{\rho} - \rho|^2 | t] \leq \min\{R^2, \mathbb{E}[e_R^2 | t]/t^2\} \leq \min\{R^2, \sigma^2 t^{2\nu-2}\}$; integrate. With $t_* = (\sigma/R)^{1/(1-\nu)}$, $\mathbb{E}_P \min\{R^2, \sigma^2 t^{2\nu-2}\} = R^2 F_t(t_*) + \sigma^2 \mathbb{E}_P[t^{2\nu-2} \mathbf{1}\{t > t_*\}]$; the first term is at most $R^2 A t_*^\beta = O(\sigma^{\beta/(1-\nu)})$ once $t_* \leq u_0$, and $\mathbb{E}_P[t^{2\nu-2} \mathbf{1}\{t > t_*\}] = \int_0^{t_*^{2\nu-2}} \mathbb{P}(t_* < t < v^{-1/(2-2\nu)}) dv \leq u_0^{2\nu-2} + A \int_{u_0^{2\nu-2}}^{t_*^{2\nu-2}} v^{-\beta/(2-2\nu)} dv$, which is $O(1)$ for $\beta > 2(1-\nu)$, $O(\log(1/\sigma))$ for $\beta = 2(1-\nu)$ and $O(t_*^{\beta-2(1-\nu)})$ for $\beta < 2(1-\nu)$. At $\nu = 1$ the bound is σ^2 before any tail enters. In the model, $c^{-2} \mathbb{E}_P |\bar{\rho} - \rho|^2 = \mathbb{E}_P \min\{1, (\sigma/c)^2 t^{2\nu-2}\}$, the same integral with equality. $\square$

Proof of Proposition 9. $\tilde{H} = H^{1/2}(I + B)H^{1/2}$ gives $\tilde{H}^{-1} = H^{-1/2}QH^{-1/2}$ and $Q = I - QB$. Hence $\tilde{k}_x^\top \tilde{H}^{-1}(Y - \tilde{M}) - k_x^\top H^{-1}(Y - M) = \tilde{g}^\top Q(W - U) - g^\top W = \tilde{g}^\top W - \tilde{g}^\top QBW - \tilde{g}^\top QU - g^\top W = v^\top W - (Q\tilde{g})^\top(BW + U)$, which is (i) since Q is symmetric; (ii) bounds its two terms. For (iii) write the same difference as $v^\top QW + g^\top(Q - I)W - \tilde{g}^\top QU$ and use $\|Q\|_{\text{op}} = \gamma_Q$ and $\|Q - I\|_{\text{op}} = \|QB\|_{\text{op}} = \eta_Q$. For (iv), $v = 0$, $U = 0$ and $B = sH^{-1}$ has eigenvalues $s/\lambda_i(H) \geq 0$, so $\eta_Q = \max_i(s/\lambda_i)/(1 + s/\lambda_i) = s/(\lambda_{\min}(H) + s)$. $\square$

B Extended diagnostic tables

The following tables preserve the detailed perturbation, ridge, synthetic and external comparison results without compressing long tables onto a single page. The public record inventory and regeneration limitations are documented separately. Numerical bounds are distinguished from approximation tolerances, and fitting objectives from reporting metrics.

Table 15: Same-data perturbation diagnostics for Proposition 9. Movement is a Euclidean prediction change in retained coordinates. The four ratio columns are *medians of entrywise bound/movement ratios*, not ratios of medians. The fourteen state-input rows are retained historical values whose JSON is absent. The feature rows are regenerated from four component records at seed 101 and four at seed 104, each with 4,000 fitting and 1,000 query rows. The signed minimum eigenvalue is rounded: a displayed -1.000 does not assert exact equality to -1 . A dash means the corresponding bound is not applicable.

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
state (archive)	Nyström 50	-0.941	0.941	2.01	57.3	4.05e+03	4.05e+03	—
state (archive)	Nyström 100	-0.872	0.872	1.05	82.1	3.05e+03	3.05e+03	—
state (archive)	Nyström 200	-0.642	0.642	0.498	136	1.65e+03	1.65e+03	—
state (archive)	Nyström 400	-0.411	0.411	0.2	178	1.54e+03	1.54e+03	—
state (archive)	Nyström 800	-0.193	0.193	0.0702	264	1.47e+03	1.47e+03	—
state (archive)	Nyström 1600	-0.078	0.078	0.0103	408	3.51e+03	3.51e+03	—
state (archive)	ridge $\times 0.5$	-0.500	0.500	0.304	517	1.34e+03	1.34e+03	—
state (archive)	ridge $\times 2.0$	+0.000	0.999	0.372	835	550	1.32e+06	550
state (archive)	ridge $\times 10.0$	+0.002	8.993	1.36	1.98e+03	271	—	271
state (archive)	ridge $\times 100.0$	+0.018	98.918	3.67	7.38e+03	111	—	111
state (archive)	PCA inputs 2	-0.998	3.600	10.5	14.3	4.81e+04	—	—
state (archive)	PCA inputs 3	-0.998	2.523	9.8	14.2	4.49e+04	—	—
state (archive)	PCA inputs 4	-0.998	1.581	5.24	25.3	6.52e+04	—	—
state (archive)	PCA inputs 5	-0.997	0.997	1.9	64.3	1.39e+05	1.39e+05	—
s101, Y_1	PCA features 1000	-0.192	0.192	0.00383	4.01e+03	3.8e+04	3.8e+04	—
s101, Y_1	PCA features 500	-0.503	0.503	0.0107	4.41e+03	5.82e+04	5.82e+04	—
s101, Y_1	PCA features 250	-0.764	0.764	0.0185	4.3e+03	1.1e+05	1.1e+05	—
s101, Y_1	PCA features 125	-0.917	0.917	0.0279	3.88e+03	2.6e+05	2.6e+05	—
s101, Y_1	PCA features 64	-0.965	0.965	0.0407	3.55e+03	4.78e+05	4.78e+05	—
s101, Y_1	PCA features 32	-0.991	0.991	0.0595	3.13e+03	1.59e+06	1.59e+06	—
s101, Y_1	PCA features 16	-0.999	0.999	0.0923	2.48e+03	1.3e+07	1.3e+07	—
s101, Y_1	Nyström 100	-1.000	1.000	0.465	637	8.85e+07	8.85e+07	—
s101, Y_1	Nyström 400	-1.000	1.000	0.223	1.27e+03	7.11e+07	7.11e+07	—
s101, Y_1	Nyström 1600	-1.000	1.000	0.0746	2.58e+03	2.74e+07	2.74e+07	—
s101, Y_1	Nyström 3200	-0.997	0.997	0.0196	5.22e+03	9.75e+06	9.75e+06	—
s101, Y_1	ridge $\times 0.5$	-0.490	0.490	0.0102	9.61e+03	5.74e+04	5.74e+04	—
s101, Y_1	ridge $\times 2.0$	+0.000	0.979	0.0115	1.7e+04	2.62e+04	2.5e+06	2.62e+04
s101, Y_1	ridge $\times 10.0$	+0.000	8.813	0.0407	4.32e+04	1.34e+04	—	1.34e+04
s101, Y_1	ridge $\times 100.0$	+0.000	96.942	0.0889	2.17e+05	6.78e+03	—	6.78e+03
s101, Y_2	PCA features 1000	-0.122	0.122	0.00143	6.41e+03	4.95e+04	4.95e+04	—
s101, Y_2	PCA features 500	-0.418	0.418	0.00425	6.8e+03	8.59e+04	8.59e+04	—

Continued on next page

Table 15 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
s101, Y_2	PCA features 250	-0.726	0.726	0.00802	6.45e+03	1.7e+05	1.7e+05	—
s101, Y_2	PCA features 125	-0.868	0.868	0.0117	6.33e+03	2.99e+05	2.99e+05	—
s101, Y_2	PCA features 64	-0.977	0.977	0.0184	5.76e+03	1.3e+06	1.3e+06	—
s101, Y_2	PCA features 32	-0.989	0.989	0.0267	5.78e+03	2.3e+06	2.3e+06	—
s101, Y_2	PCA features 16	-0.999	0.999	0.0419	4.83e+03	1.49e+07	1.49e+07	—
s101, Y_2	Nyström 100	-1.000	1.000	0.345	731	1.01e+08	1.01e+08	—
s101, Y_2	Nyström 400	-1.000	1.000	0.122	1.97e+03	8.48e+07	8.48e+07	—
s101, Y_2	Nyström 1600	-1.000	1.000	0.0823	2.38e+03	5.34e+07	5.34e+07	—
s101, Y_2	Nyström 3200	-0.998	0.998	0.0161	7.05e+03	1.86e+07	1.86e+07	—
s101, Y_2	ridge $\times 0.5$	-0.491	0.491	0.00632	1.71e+04	7.71e+04	7.71e+04	—
s101, Y_2	ridge $\times 2.0$	+0.000	0.982	0.00687	3.14e+04	3.64e+04	3.96e+06	3.64e+04
s101, Y_2	ridge $\times 10.0$	+0.000	8.836	0.0249	7.8e+04	1.82e+04	—	1.82e+04
s101, Y_2	ridge $\times 100.0$	+0.000	97.199	0.0545	3.92e+05	9.17e+03	—	9.17e+03
s101, Y_3	PCA features 1000	-0.170	0.170	0.00292	6.17e+03	4.56e+04	4.56e+04	—
s101, Y_3	PCA features 500	-0.475	0.475	0.00766	7.02e+03	7.73e+04	7.73e+04	—
s101, Y_3	PCA features 250	-0.738	0.738	0.0136	6.44e+03	1.38e+05	1.38e+05	—
s101, Y_3	PCA features 125	-0.898	0.898	0.0211	5.61e+03	2.87e+05	2.87e+05	—
s101, Y_3	PCA features 64	-0.964	0.964	0.028	5.45e+03	7.06e+05	7.06e+05	—
s101, Y_3	PCA features 32	-0.990	0.990	0.0406	4.9e+03	2.09e+06	2.09e+06	—
s101, Y_3	PCA features 16	-0.999	0.999	0.0725	3.51e+03	1.63e+07	1.63e+07	—
s101, Y_3	Nyström 100	-1.000	1.000	0.35	876	4.97e+07	4.97e+07	—
s101, Y_3	Nyström 400	-1.000	1.000	0.191	1.55e+03	4.53e+07	4.53e+07	—
s101, Y_3	Nyström 1600	-0.999	0.999	0.0709	3.33e+03	1.45e+07	1.45e+07	—
s101, Y_3	Nyström 3200	-0.997	0.997	0.0331	4.54e+03	6.39e+06	6.39e+06	—
s101, Y_3	ridge $\times 0.5$	-0.492	0.492	0.00865	1.65e+04	7.26e+04	7.26e+04	—
s101, Y_3	ridge $\times 2.0$	+0.000	0.985	0.00949	3e+04	3.38e+04	4.43e+06	3.38e+04
s101, Y_3	ridge $\times 10.0$	+0.000	8.864	0.0357	7.18e+04	1.63e+04	—	1.63e+04
s101, Y_3	ridge $\times 100.0$	+0.000	97.500	0.0821	3.44e+05	7.81e+03	—	7.81e+03
s101, Y_4	PCA features 1000	-0.006	0.006	0.00138	538	4.64e+03	4.64e+03	—
s101, Y_4	PCA features 500	-0.032	0.032	0.00463	712	7.83e+03	7.83e+03	—
s101, Y_4	PCA features 250	-0.114	0.114	0.0104	826	1.35e+04	1.35e+04	—
s101, Y_4	PCA features 125	-0.269	0.269	0.0191	897	2.12e+04	2.12e+04	—
s101, Y_4	PCA features 64	-0.594	0.594	0.0286	963	5.8e+04	5.8e+04	—
s101, Y_4	PCA features 32	-0.741	0.741	0.0436	983	8.06e+04	8.06e+04	—
s101, Y_4	PCA features 16	-0.902	0.902	0.0769	987	1.65e+05	1.65e+05	—
s101, Y_4	Nyström 100	-0.998	0.998	0.603	304	1.2e+06	1.2e+06	—

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Table 15 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
s101, Y_4	Nyström 400	-0.985	0.985	0.35	500	2.11e+05	2.11e+05	–
s101, Y_4	Nyström 1600	-0.860	0.860	0.135	916	4.86e+04	4.86e+04	–
s101, Y_4	Nyström 3200	-0.762	0.762	0.0397	1.71e+03	8.46e+04	8.46e+04	–
s101, Y_4	ridge $\times 0.5$	-0.494	0.494	0.063	3.15e+03	1.61e+04	1.61e+04	–
s101, Y_4	ridge $\times 2.0$	+0.000	0.989	0.0599	6.6e+03	8.63e+03	1.56e+06	8.63e+03
s101, Y_4	ridge $\times 10.0$	+0.000	8.901	0.191	1.86e+04	4.89e+03	–	4.89e+03
s101, Y_4	ridge $\times 100.0$	+0.000	97.908	0.36	1.07e+05	2.86e+03	–	2.86e+03
s104, Y_1	PCA features 1000	-0.184	0.184	0.00397	4.68e+03	3.85e+04	3.85e+04	–
s104, Y_1	PCA features 500	-0.471	0.471	0.0104	5.01e+03	5.82e+04	5.82e+04	–
s104, Y_1	PCA features 250	-0.724	0.724	0.0175	4.8e+03	1.04e+05	1.04e+05	–
s104, Y_1	PCA features 125	-0.901	0.901	0.0267	4.25e+03	2.45e+05	2.45e+05	–
s104, Y_1	PCA features 64	-0.959	0.959	0.0373	3.87e+03	4.76e+05	4.76e+05	–
s104, Y_1	PCA features 32	-0.985	0.985	0.055	3.43e+03	1.07e+06	1.07e+06	–
s104, Y_1	PCA features 16	-0.997	0.997	0.0893	2.61e+03	3.51e+06	3.51e+06	–
s104, Y_1	concatenation	-0.749	76.009	0.0427	2.3e+05	1.99e+05	–	–
s104, Y_1	member 1	-0.973	175.948	0.049	2.34e+05	2.24e+06	–	–
s104, Y_1	member 2	-0.967	126.717	0.0541	3.81e+05	2.02e+06	–	–
s104, Y_1	Nyström 100	-1.000	1.000	0.861	379	9.61e+07	9.61e+07	–
s104, Y_1	Nyström 400	-1.000	1.000	0.247	1.25e+03	6.69e+07	6.69e+07	–
s104, Y_1	Nyström 1600	-1.000	1.000	0.112	2.01e+03	4.7e+07	4.7e+07	–
s104, Y_1	Nyström 3200	-1.000	1.000	0.0249	4.43e+03	5.86e+07	5.86e+07	–
s104, Y_1	ridge $\times 0.5$	-0.493	0.493	0.011	1.06e+04	5.96e+04	5.96e+04	–
s104, Y_1	ridge $\times 2.0$	+0.000	0.986	0.0124	1.89e+04	2.7e+04	3.93e+06	2.7e+04
s104, Y_1	ridge $\times 10.0$	+0.000	8.877	0.0445	4.73e+04	1.36e+04	–	1.36e+04
s104, Y_1	ridge $\times 100.0$	+0.000	97.646	0.102	2.27e+05	6.54e+03	–	6.54e+03
s104, Y_2	PCA features 1000	-0.108	0.108	0.00163	7.72e+03	5.02e+04	5.02e+04	–
s104, Y_2	PCA features 500	-0.370	0.370	0.00553	8.84e+03	7.23e+04	7.23e+04	–
s104, Y_2	PCA features 250	-0.668	0.668	0.0107	8.77e+03	1.29e+05	1.29e+05	–
s104, Y_2	PCA features 125	-0.834	0.834	0.0166	7.71e+03	2.12e+05	2.12e+05	–
s104, Y_2	PCA features 64	-0.955	0.955	0.0234	6.59e+03	6.62e+05	6.62e+05	–
s104, Y_2	PCA features 32	-0.983	0.983	0.0326	6.12e+03	1.46e+06	1.46e+06	–
s104, Y_2	PCA features 16	-0.998	0.998	0.0503	5.04e+03	1.05e+07	1.05e+07	–
s104, Y_2	concatenation	-0.734	96.255	0.0279	6.7e+05	3.68e+05	–	–
s104, Y_2	member 1	-0.974	200.331	0.0328	8.4e+05	5.59e+06	–	–
s104, Y_2	member 2	-0.968	266.674	0.035	9.16e+05	3.4e+06	–	–
s104, Y_2	Nyström 100	-1.000	1.000	0.713	451	1.85e+08	1.85e+08	–

Continued on next page

Table 15 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
s104, Y_2	Nyström 400	-1.000	1.000	0.119	2.5e+03	4.07e+07	4.07e+07	–
s104, Y_2	Nyström 1600	-0.999	0.999	0.0322	6.59e+03	1.8e+07	1.8e+07	–
s104, Y_2	Nyström 3200	-0.992	0.992	0.0104	1.25e+04	8.1e+06	8.1e+06	–
s104, Y_2	ridge $\times 0.5$	-0.498	0.498	0.00788	2.13e+04	8.49e+04	8.49e+04	–
s104, Y_2	ridge $\times 2.0$	+0.000	0.995	0.00862	3.89e+04	3.91e+04	1.66e+07	3.91e+04
s104, Y_2	ridge $\times 10.0$	+0.000	8.958	0.0323	9.36e+04	1.88e+04	–	1.88e+04
s104, Y_2	ridge $\times 100.0$	+0.000	98.534	0.0733	4.53e+05	9.13e+03	–	9.13e+03
s104, Y_3	PCA features 1000	-0.176	0.176	0.00305	6.26e+03	4.48e+04	4.48e+04	–
s104, Y_3	PCA features 500	-0.474	0.474	0.00801	7.06e+03	7.26e+04	7.26e+04	–
s104, Y_3	PCA features 250	-0.738	0.738	0.0145	6.18e+03	1.26e+05	1.26e+05	–
s104, Y_3	PCA features 125	-0.882	0.882	0.0189	6.1e+03	2.67e+05	2.67e+05	–
s104, Y_3	PCA features 64	-0.957	0.957	0.0259	5.75e+03	6.3e+05	6.3e+05	–
s104, Y_3	PCA features 32	-0.990	0.990	0.0371	5.12e+03	2.16e+06	2.16e+06	–
s104, Y_3	PCA features 16	-0.999	0.999	0.0648	3.65e+03	2.51e+07	2.51e+07	–
s104, Y_3	concatenation	-0.749	12.229	0.0228	6.41e+04	2.52e+05	–	–
s104, Y_3	member 1	-0.987	28.650	0.0374	1.01e+05	5.18e+06	–	–
s104, Y_3	member 2	-0.978	32.067	0.0391	3.02e+04	2.59e+06	–	–
s104, Y_3	Nyström 100	-1.000	1.000	0.777	389	1.27e+08	1.27e+08	–
s104, Y_3	Nyström 400	-1.000	1.000	0.203	1.42e+03	5.4e+07	5.4e+07	–
s104, Y_3	Nyström 1600	-1.000	1.000	0.134	1.72e+03	4.38e+07	4.38e+07	–
s104, Y_3	Nyström 3200	-0.999	0.999	0.029	4.4e+03	1.58e+07	1.58e+07	–
s104, Y_3	ridge $\times 0.5$	-0.487	0.487	0.00833	1.62e+04	7.25e+04	7.25e+04	–
s104, Y_3	ridge $\times 2.0$	+0.000	0.975	0.00917	2.93e+04	3.42e+04	2.65e+06	3.42e+04
s104, Y_3	ridge $\times 10.0$	+0.000	8.771	0.0329	7.36e+04	1.73e+04	–	1.73e+04
s104, Y_3	ridge $\times 100.0$	+0.000	96.479	0.0765	3.48e+05	8.23e+03	–	8.23e+03
s104, Y_4	PCA features 1000	-0.003	0.003	0.000765	496	4.96e+03	4.96e+03	–
s104, Y_4	PCA features 500	-0.019	0.019	0.00303	682	8.81e+03	8.81e+03	–
s104, Y_4	PCA features 250	-0.088	0.088	0.00823	778	1.58e+04	1.58e+04	–
s104, Y_4	PCA features 125	-0.259	0.259	0.0153	851	3.11e+04	3.11e+04	–
s104, Y_4	PCA features 64	-0.406	0.406	0.0238	906	4e+04	4e+04	–
s104, Y_4	PCA features 32	-0.790	0.790	0.0427	859	1.27e+05	1.27e+05	–
s104, Y_4	PCA features 16	-0.883	0.883	0.0733	851	1.62e+05	1.62e+05	–
s104, Y_4	concatenation	-0.658	58.594	0.0605	1.49e+04	1.97e+05	–	–
s104, Y_4	member 1	-0.897	23.688	0.103	5.19e+03	3.18e+05	–	–
s104, Y_4	member 2	-0.919	155.529	0.108	1.74e+04	8.07e+05	–	–
s104, Y_4	Nyström 100	-0.993	0.993	0.517	378	3.93e+05	3.93e+05	–

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Table 15 (continued)

source	perturbation	$\lambda_{\min}(B)$	$\ B\ $	median movement	residual-action ratio	signed-spectrum ratio	$\ B\ < 1$ ratio	ridge ratio
s104, Y_4	Nyström 400	-0.961	0.961	0.241	772	1.39e+05	1.39e+05	–
s104, Y_4	Nyström 1600	-0.850	0.850	0.0775	1.66e+03	9.85e+04	9.85e+04	–
s104, Y_4	Nyström 3200	-0.707	0.707	0.0213	2.57e+03	1.5e+05	1.5e+05	–
s104, Y_4	ridge $\times 0.5$	-0.491	0.491	0.0482	4.89e+03	2.63e+04	2.63e+04	–
s104, Y_4	ridge $\times 2.0$	+0.000	0.983	0.0525	8.97e+03	1.24e+04	1.43e+06	1.24e+04
s104, Y_4	ridge $\times 10.0$	+0.000	8.846	0.179	2.36e+04	6.6e+03	–	6.6e+03
s104, Y_4	ridge $\times 100.0$	+0.000	97.301	0.393	1.18e+05	3.32e+03	–	3.32e+03

Table 16: Ridge-path diagnostics, seed 101, 4,000 fitting and 1,000 query rows. The bias and noise factor refer to Proposition 4(iii); the factor multiplies a hypothetical noise variance and is not an estimate of simulator noise. Corrected and kernel-only test columns are mean relative errors of a physical spectral component, in percent. The final column uses retained PCA coordinates on the held-out half and compares same-half fitting with split-half fitting; it is a different metric and sample. Each ridge is evaluated descriptively, not selected on these test values.

component	kernel	γ	$\text{tr}S_\lambda$	bias	noise factor	median $\ a_\lambda\ ^2$	median $\tilde{P}_\lambda$	corrected [%]	kernel only [%]	same / split half PCA error [%]
Y1	input	1e-08	3841	0.000226	0.924	0.87	0.0324	0.2168	0.1754	0.079 / 2.465
Y1	input	1e-06	1624	0.0712	0.26	0.226	0.0395	0.2376	0.2420	1.145 / 2.214
Y1	input	0.0001	212	0.282	0.0269	0.0237	0.0997	0.4132	0.9768	2.221 / 2.395
Y1	input	0.01	19	0.413	0.00194	0.00197	0.315	0.5095	5.7578	2.716 / 2.728
Y1	feature	1e-08	2541	0.0012	0.51	0.481	0.00637	0.2027	0.2111	0.280 / 1.171
Y1	feature	1e-06	689	0.0225	0.111	0.0668	0.0151	0.3130	0.3275	1.085 / 1.464
Y1	feature	0.0001	103	0.156	0.0137	0.00825	0.0529	0.4546	0.6396	2.105 / 2.187
Y1	feature	0.01	9	0.418	0.001	0.000845	0.185	0.5326	5.9929	2.783 / 2.786
Y2	input	1e-08	3884	0.000107	0.943	0.523	0.0369	0.4559	0.6885	0.043 / 1.784
Y2	input	1e-06	1554	0.061	0.219	0.158	0.0416	0.4849	0.9183	0.886 / 1.632
Y2	input	0.0001	132	0.211	0.0143	0.0127	0.0845	0.6325	4.4719	1.516 / 1.580
Y2	input	0.01	10	0.273	0.00115	0.00117	0.227	0.8581	26.9965	1.773 / 1.777
Y2	feature	1e-08	2339	0.000906	0.451	0.44	0.00562	0.1431	0.1439	0.184 / 0.738
Y2	feature	1e-06	564	0.0159	0.0893	0.0618	0.0143	0.2768	0.2696	0.662 / 0.899
Y2	feature	0.0001	86	0.134	0.0118	0.00829	0.0512	0.5764	1.0187	1.275 / 1.326
Y2	feature	0.01	9	0.27	0.000998	0.000909	0.188	0.8905	15.5377	1.738 / 1.741
Y3	input	1e-08	3308	0.00797	0.7	0.412	0.0132	0.2382	0.3731	0.324 / 2.147
Y3	input	1e-06	558	0.283	0.0644	0.0523	0.0192	0.2627	0.9135	1.656 / 1.951
Y3	input	0.0001	45	0.455	0.00546	0.00501	0.0457	0.3432	5.4172	2.173 / 2.155
Y3	input	0.01	5	0.554	0.000642	0.000643	0.152	0.4914	19.3748	2.715 / 2.702
Y3	feature	1e-08	2371	0.00158	0.461	0.458	0.00569	0.0999	0.1197	0.221 / 0.903
Y3	feature	1e-06	590	0.0255	0.094	0.063	0.0145	0.1650	0.1866	0.829 / 1.136
Y3	feature	0.0001	90	0.267	0.0123	0.00842	0.0513	0.3145	0.6620	1.790 / 1.882
Y3	feature	0.01	9	0.525	0.001	0.000885	0.189	0.4913	7.4936	2.612 / 2.624
Y4	input	1e-08	3998	9.17e-07	0.999	0.473	0.269	2.1826	2.3706	0.007 / 12.844
Y4	input	1e-06	3792	0.00736	0.9	0.437	0.27	2.1557	2.3495	0.629 / 12.779
Y4	input	0.0001	1074	1.16	0.125	0.0912	0.319	1.9633	2.4907	7.986 / 11.573
Y4	input	0.01	51	2.3	0.00312	0.00306	0.619	2.3401	8.9051	12.340 / 12.554
Y4	feature	1e-08	3823	0.00807	0.917	0.975	0.031	2.4285	2.4310	0.626 / 14.459
Y4	feature	1e-06	1652	0.794	0.272	0.24	0.0392	1.8004	1.8078	6.037 / 11.377
Y4	feature	0.0001	229	1.83	0.0293	0.0257	0.103	1.8454	1.9424	9.636 / 10.497
Y4	feature	0.01	19	2.39	0.0019	0.00181	0.323	2.4706	4.7750	12.907 / 12.992

β	average-family measured slope	$\beta/(\beta + 2)$	uniform-family measured slope	analytic slope same ladder
0.5	0.200	0.200	0.494	0.494
1	0.333	0.333	0.973	0.973
2	0.498	0.500	1.721	1.724
3	0.597	0.600	1.963	1.963
4	0.664	0.667	1.995	1.995

β	uniform-family maximum relative gap	profile-family maximum relative gap	violations
0.5	5.7e-04	4.4e-04	0
1	1.7e-03	3.9e-04	0
2	1.0e-02	3.1e-04	0
3	1.4e-03	5.2e-04	0
4	2.8e-04	2.8e-04	0

Table 17: Manufactured transfer families, regenerated with seed zero and two million samples for each β . The first panel compares finite-ladder slopes, including the analytic curve on that same ladder. The second reports maximum relative discrepancies between Monte Carlo MSE and the uniform-family analytic or profile-family quadrature reference, and total pointwise violation counts. These are numerical checks of the construction, not fitted EMIT rates. The source JSON also records rare-event counts and the points retained in each average-family slope fit.

Table 18: Matched correction-coefficient benchmark [4]. The standard split uses 35,000 / 7,500 / 7,500 states. OOD test states exceed the 0.85 aerosol or water-vapor quantile; remaining states are split 85/15. Cells average three seeds or initializations; published reference rows are not rerun. RMSE, MAE and SMAPE are pooled; R^2 is uniformly averaged across the three coefficients, reconstructed and checked from archived summary identities. Model keys: dkr, feature kernel; krr, input kernel; mlp, network; ens, ensemble; resid, residual correction; lowfi, the low-fidelity mean. The categorical inputs follow the benchmark protocol.

split	inputs	head	RMSE	MAE	R^2	SMAPE [%]
OOD	state only	dkr	0.01408±0.00177	0.00709±0.00083	0.99226±0.00106	5.714±0.441
OOD	state only	krr	0.11283±0.00069	0.05430±0.00197	0.66968±0.01490	30.762±1.160
OOD	state only	mlp	0.00750±0.00044	0.00355±0.00036	0.99586±0.00012	4.500±0.249
OOD	state only	mlp_ens	0.00624±0.00015	0.00258±0.00017	0.99667±0.00004	3.470±0.370
OOD	state only	mlp_resid	0.00620±0.00015	0.00252±0.00015	0.99670±0.00005	3.372±0.308
OOD	state only	ridge	0.11944±0.00000	0.06195±0.00000	0.49009±0.00000	37.228±0.000
OOD	with 6S inputs	dkr	0.00735±0.00036	0.00336±0.00033	0.99597±0.00022	4.101±0.474
OOD	with 6S inputs	krr	0.02904±0.00015	0.01432±0.00005	0.97160±0.00020	11.238±0.033
OOD	with 6S inputs	lowfi_resid	0.02901±0.00016	0.01430±0.00006	0.97172±0.00021	11.416±0.047
OOD	with 6S inputs	mlp	0.00741±0.00080	0.00347±0.00063	0.99602±0.00048	4.406±0.366
OOD	with 6S inputs	mlp_ens	0.00611±0.00030	0.00246±0.00023	0.99682±0.00013	3.009±0.093
OOD	with 6S inputs	mlp_resid	0.00596±0.00031	0.00232±0.00022	0.99687±0.00012	2.921±0.067
OOD	with 6S inputs	ridge	0.08048±0.00000	0.04396±0.00000	0.79274±0.00000	29.064±0.000
standard	state only	dkr	0.00874±0.00121	0.00387±0.00071	0.99379±0.00053	5.342±0.323
standard	state only	krr	0.11288±0.00001	0.05157±0.00002	0.68464±0.00028	31.178±0.032
standard	state only	mlp	0.00657±0.00026	0.00242±0.00030	0.99465±0.00007	5.136±0.550
standard	state only	mlp_ens	0.00591±0.00007	0.00170±0.00005	0.99513±0.00010	3.873±0.132
standard	state only	mlp_resid	0.00581±0.00004	0.00151±0.00004	0.99520±0.00010	3.543±0.066
standard	state only	ridge	0.11959±0.00000	0.06067±0.00000	0.48572±0.00000	41.647±0.000
standard	with 6S inputs	dkr	0.00627±0.00025	0.00218±0.00021	0.99502±0.00028	4.429±0.169
standard	with 6S inputs	krr	0.02514±0.00010	0.01155±0.00003	0.97447±0.00010	10.184±0.037
standard	with 6S inputs	lowfi_resid	0.02513±0.00010	0.01155±0.00003	0.97452±0.00010	10.178±0.041
standard	with 6S inputs	mlp	0.00632±0.00018	0.00224±0.00017	0.99504±0.00014	4.496±0.160
standard	with 6S inputs	mlp_ens	0.00587±0.00014	0.00163±0.00012	0.99531±0.00011	3.407±0.130
standard	with 6S inputs	mlp_resid	0.00578±0.00012	0.00148±0.00008	0.99535±0.00010	3.185±0.097
standard	with 6S inputs	ridge	0.06981±0.00000	0.03631±0.00000	0.83813±0.00000	28.300±0.000
standard	published	pKANrtm	0.00619	0.00186	0.99472	3.912
OOD	published	pKANrtm	0.01022	0.00544	0.99418	5.399

Table 19: OCO-2 selection comparison, seeds 0–2 on each of three bands. Each record uses 2,000 validation and 2,000 test rows. Both reported columns are *mean relative norms* in percent; the stack fitting objectives are row-relative *squared* errors. The radiance quadratic uses the exact affine reconstruction Gram matrix. Model keys: dkr, feature kernel; mean, trained network; ridge, linear readout; raw/ard, input kernels. Flat, radx and wnum distinguish the three archived training losses. Selection is per coordinate, with the radiance variant using greedy coordinate updates, not exhaustive global search.

band	head / rule	reduced error [%]	radiance error [%]
o2	select_radiance	4.08±0.05	0.0283±0.0017
o2	select_reduced	4.09±0.06	0.0283±0.0017
o2	single dkr_flat	4.07±0.06	0.0757±0.0012
o2	single dkr_radx	20.14±0.07	0.0351±0.0006
o2	single dkr_wnum	21.70±0.60	0.0309±0.0017
o2	single kernel_ard	30.02±0.03	0.2706±0.1030
o2	single kernel_raw	40.53±0.09	0.2353±0.0047
o2	single mean_flat	4.40±0.08	0.1756±0.0111
o2	single mean_radx	58.72±0.60	0.0510±0.0049
o2	single mean_wnum	49.22±0.34	0.0437±0.0002
o2	single ridge_flat	4.67±0.09	0.2205±0.0124
o2	single ridge_radx	44.83±0.43	0.0538±0.0044
o2	single ridge_wnum	45.55±0.73	0.0527±0.0016
o2	stack_radiance	18.59±0.25	0.0264±0.0007
o2	stack_reduced	4.10±0.07	0.0291±0.0029
sco2	select_radiance	8.09±0.02	0.0598±0.0013
sco2	select_reduced	8.09±0.02	0.0598±0.0013
sco2	single dkr_flat	8.09±0.02	0.1083±0.0114
sco2	single dkr_radx	15.27±0.25	0.0780±0.0094
sco2	single dkr_wnum	15.48±0.31	0.0638±0.0022
sco2	single kernel_ard	28.72±0.02	0.6023±0.0097
sco2	single kernel_raw	41.01±0.09	0.5229±0.0081
sco2	single mean_flat	8.23±0.01	0.2107±0.0194
sco2	single mean_radx	46.08±0.91	0.0936±0.0082
sco2	single mean_wnum	36.64±0.38	0.0750±0.0020
sco2	single ridge_flat	8.34±0.02	0.2568±0.0244
sco2	single ridge_radx	41.14±1.78	0.0967±0.0081
sco2	single ridge_wnum	40.25±0.24	0.0887±0.0034
sco2	stack_radiance	8.87±0.15	0.0511±0.0016
sco2	stack_reduced	8.09±0.01	0.0536±0.0020
wco2	select_radiance	16.26±0.07	0.0510±0.0086
wco2	select_reduced	16.26±0.07	0.0510±0.0086
wco2	single dkr_flat	16.26±0.07	0.1135±0.0069
wco2	single dkr_radx	48.28±0.28	0.0475±0.0055
wco2	single dkr_wnum	48.60±0.38	0.0516±0.0086
wco2	single kernel_ard	52.45±0.07	0.2231±0.0014
wco2	single kernel_raw	58.63±0.20	0.4431±0.0063
wco2	single mean_flat	16.42±0.06	0.2254±0.0161
wco2	single mean_radx	67.92±0.46	0.0605±0.0075
wco2	single mean_wnum	62.39±0.42	0.0700±0.0087
wco2	single ridge_flat	16.49±0.07	0.2571±0.0234
wco2	single ridge_radx	61.03±0.31	0.0663±0.0080
wco2	single ridge_wnum	61.33±0.05	0.0778±0.0095
wco2	stack_radiance	37.43±0.20	0.0391±0.0026
wco2	stack_reduced	16.28±0.07	0.0404±0.0034

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital; both entities supported this work and are named above. These entities and the public funders had no role in the study design, analysis, manuscript preparation, or decision to publish.

AI-assisted contributions

Claude Code (Anthropic) and Codex (OpenAI) implemented and ran the data pipeline, model comparisons, scaling experiments, diagnostics and figure generation, and drafted the manuscript from the author’s data and the methodology of [12]. ChatGPT (OpenAI) developed the residual-error and inversion-stability arguments, checked the finite identities, analysed the stored per-split results, implemented the conditioning diagnostics and verification tests, and edited the manuscript. Claude Code implemented and ran the later experiments of Sections 9–12: the transmission-conditioned retrieval diagnostics, the ten-kernel comparison and its combiners, the regularisation suite, the ridge-path and perturbation measurements, the synthetic reference, the matched benchmark rescoring, and the OCO-2 selection through the Gram matrix of its design; it also generated every table of those sections from the stored run records and wrote their text. The author supplied the data, set the questions and reviewed the results. The present revision used ChatGPT to reconcile the manuscript with the archived implementation, separate precision cohorts, add synthetic regression tests, and rebuild the manuscript; this work did not rerun the trained-model campaigns.

Data and code availability

The manuscript, model drivers, run summaries and reconstruction scripts are available at <https://github.com/yspennstate/emit-oco2-kernel-emulators>. The reproduction documentation distinguishes rebuilding a table from recorded metrics, rerunning a synthetic check, and reproducing model training from raw data.

The main width-512 and width-2000 EMIT comparisons each have ten per-split records with array digests, hyperparameters, split sizes and evaluation metrics. These twenty records reproduce Tables 1, 7, 2 and 3. The hashes identify recorded arrays; regenerating these tables does not independently verify the unavailable arrays or retrain their models. Secondary learning-size, width and tuning experiments retain ten-run aggregate summaries but incomplete individual archives: one isotropic and one input-scaled 8,000-row record are absent, width 1000 has three

individual records, and full-data tuning has five. The original three-member experiment uses three splits and the width-4000 experiment two.

For the later diagnostics, four float64 transmission records at seeds 104–107 are summarized together; the separate seed-101 float32 record is identified as a precision-altered experiment and is not pooled into that comparison. The public archive also contains twelve multi-kernel component/seed records, three regularization records, eight feature-perturbation records, four ridge-path records, nine OCO-2 selection records and the matched-benchmark scoring records. The four-split concatenation table and the joint-stacking table are retained historical aggregates: their generating campaign and stack JSON files are absent from the public archive. The state-input part of the perturbation table likewise lacks its generating JSON, whereas the feature rows can be regenerated. These distinctions are recorded by table label in the reproduction manifest; missing records are not replaced by smaller or differently defined experiments.

The EMIT arrays, identified as `data.EMIT_24k` and attributed to the Jet Propulsion Laboratory, are not redistributed. The author offers raw arrays, trained weights and prediction files on request. The public archive does not establish the simulator version, generation configuration or redistribution permissions. These remain requirements for independent reproduction of the training campaign. Scoring in float64 does not recover information lost when a prediction was stored in float32. Precision-altered inputs therefore require an explicit opt-in in the diagnostic drivers. The primary comparisons are all-band scores; supplementary common-mask and transmission diagnostics do not constitute a validated operational retrieval protocol.

The OCO-2 arrays and reference predictions are from the release accompanying [3]. The other configurations use the cited releases. No new trained EMIT, OCO-2 or multi-fidelity model is introduced by the present manuscript revision.

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